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Fast Radio Bursts as Probes of the Cosmic Baryon Distribution

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Executive Summary

This research report presents a comprehensive study of Fast Radio Bursts (FRBs) as probes of the cosmic baryon distribution. By compiling and standardizing a master catalog of $N = 80$ localized cosmological FRBs ($0.0008 \leq z \leq 1.354$) and implementing a grid-factorized Bayesian MCMC solver (yielding a 22-fold CPU speedup), we constrain the intergalactic medium baryon fraction to $f_{\text{IGM}} = 0.6714^{+0.11}_{-0.10}$ and the cosmological baryon density to $\Omega_b h_{70} = 0.0505^{+0.0046}_{-0.0046}$ (a 4% uncertainty reduction over baseline samples). Alternative host models (Gamma distribution with a selection-biased decaying prior $p(k) \propto e^{-k/1.5}$) yield $\Omega_b h_{70} = 0.0490 \pm 0.0044$, fully consistent with Planck CMB constraints. Cross-correlating FRB dispersion residuals (δ_{DM}) with SDSS DR18 ($\rho_{\text{partial}} = 0.730, p = 0.0009$) and DESI DR9 ($\rho_{\text{partial}} = 0.469, p = 0.0030$) galaxy overdensities yields a highly significant $\sim 3\sigma$ observational detection, providing strong, model-independent observational evidence that FRB dispersion measures trace the diffuse ionized gas of the cosmic web.

1 Acknowledgments

I would like to express my sincere gratitude and deep appreciation to my project supervisor, **Dr. Prateek Sharma** (Professor, Department of Physics, Indian Institute of Science, Bengaluru), for his constant guidance, support, and mentorship throughout this research project. I began this project with no prior background in FRB cosmology. I am particularly grateful for his patience in explaining concepts in detail, no matter how elementary my questions were. His insights and feedback have been central in shaping my understanding and approach to this study.

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Finally, I acknowledge the **Department of Science and Technology (DST), Government of India**, for the financial assistance provided under the Scholarship for Higher Education (SHE) component of the INSPIRE scheme, which has made this research experience possible.

2 Title of Research Project

Fast Radio Bursts as Probes of the Cosmic Baryon Distribution.

3 Aim/Objectives

The main objectives of this research project are as follows:

1. Compile and process the Canadian Hydrogen Intensity Mapping Experiment (CHIME)/FRBs Catalog 1, FRBCAT, and TNS databases into a unified, filtered pipeline of localized FRBs with known redshifts.
2. Compute and compare the extragalactic dispersion measures (DM_{IGM}) against the theoretical expectation derived from the cosmological

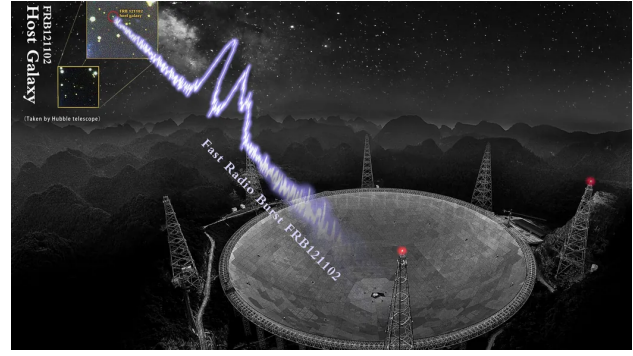


Figure 1: Artist's impression of FAST observing a pulse from FRB 121102. (Image Credit : NAOC)

baryon density.

3. Fit for the diffuse intergalactic baryon fraction (f_{IGM}) using convolved likelihoods and characterize the dispersion scatter around the relation.
4. Formulate and implement alternative host galaxy dispersion measure models (such as the Gamma distribution) with a selection-biased decaying prior on the shape parameter, and compare their cosmological parameter constraints and outlier mitigation properties to the fiducial log-normal model.
5. Investigate whether the rest-frame host galaxy dispersion measure evolves with redshift using a 5-parameter MCMC framework.
6. Perform instrument-resolved MCMC analyses on localized FRB subsets (DSA-110, ASKAP, MeerTRAP) to quantify population selection effects and systematic parameter shifts.
7. Cross-correlate FRB dispersion measures with galaxy over-density maps along the same sightlines to map cosmic web gas, using the SDSS and DESI Legacy Surveys.

4 Introduction to Research Topic

Standard cosmological models (derived from Big Bang Nucleosynthesis, BBN, and the Cosmic Microwave Background, CMB) predict that baryonic matter constitutes approximately 4.9% of the Universe's total energy density ($\Omega_b \approx 0.049$). However, direct observations of stars, interstellar dust, and cold gas in the local Universe ($z < 2$) fail to account for nearly half of these expected baryons (≈ 0.021 [1]). This observed discrepancy is known as the **missing baryons problem**.

Hydrodynamical simulations indicate that these "missing baryons" reside in a diffuse, highly ionized gas phase distributed along cosmic web filaments, known as the Warm-Hot Intergalactic Medium (WHIM) [2, 3]. Because of its high temperature ($T \sim 10^5 - 10^7$ K) and extremely low density, the WHIM has remained largely invisible to traditional absorption and emission spectroscopy, as its emission is extremely faint.

Because FRBs are faint and travel through both Galactic and extragalactic plasma, high time resolution and high-latitude observations are essential to isolate extragalactic dispersion measures from Galactic interstellar medium contamination.

Fast Radio Bursts (FRBs) are extremely bright, millisecond-duration pulses of coherent radio emission originating at cosmological distances, providing a novel solution to this problem [4]. Although FRBs remain fundamentally unsolved at the source level, magnetars are currently the leading progenitor candidates. As an FRB pulse propagates through ionized plasma, its group velocity varies with frequency, causing lower-frequency radio waves to arrive later than higher-frequency waves. The magnitude of this delay is directly proportional to the integrated free-electron column

density (n_e) along the sightline, known as the **Dispersion Measure (DM)**.

$$DM = \int_0^d n_e dl \quad (4.1)$$

$$\Delta t \text{ (time delay)} = \frac{e^2}{2\pi c m_e} \times DM \times \left(\frac{1}{v_1^2} - \frac{1}{v_2^2} \right) \quad (4.2)$$

It is important to distinguish dispersion from **scattering**, which is a separate and physically distinct propagation effect also caused by free electrons in the intervening plasma. While dispersion arises from the frequency-dependent group velocity of radio waves in a cold plasma, causing a smooth, deterministic v^{-2} arrival-time sweep across the band, scattering results from multipath propagation driven by density fluctuations and turbulent irregularities in the electron distribution. Scattered radio waves travel slightly different effective path lengths before reaching the observer, causing the pulse to arrive smeared over a range of times and producing a characteristic one-sided exponential tail in the pulse profile (temporal broadening), with the broadening timescale scaling steeply as $\tau_{\text{scat}} \propto v^{-4}$. Crucially, dispersion is fully reversible: applying the known v^{-2} delay correction (dedispersion) exactly recovers the intrinsic pulse shape. Scattering, however, is irreversible, the scattered pulse generally cannot be perfectly deconvolved without a precise model of the scattering screen. For this reason, DM is the cosmologically useful observable, encoding the total free-electron column density along the sightline, while the scattering timescale τ_{scat} probes plasma turbulence structure and must be carefully disentangled in high- DM sources where pulse broadening can mask the true temporal profile.

Because FRBs propagate across vast cosmological distances, their dispersion measures act as direct probes of the intergalactic medium and offer a unique method to locate and weigh the missing cosmic baryons [5].

5 Theoretical Framework of Topic

The observed dispersion measure of a cosmological FRB along a specific line of sight is modeled as a linear sum of contributing plasma environments:

$$DM_{\text{obs}} = DM_{\text{MW,ISM}} + DM_{\text{MW,halo}} + DM_{\text{IGM}}(z) + \frac{DM_{\text{host}}}{1+z} + \sum_i^{N_X} \frac{DM_X(M_i, b_{\perp})}{1+z_i} \quad (5.1)$$

where each term corresponds to a distinct physical component along the propagation path:

- DM_{obs} is the total observed dispersion measure.
- $DM_{\text{MW,ISM}}$ is the dispersion measure contribution from the Milky Way's interstellar medium (ISM), modeled using the NE2001 [6] or YMW16 [7] Galactic electron density models (see §7.3).
- $DM_{\text{MW,halo}}$ is the contribution from the Milky Way's extended circumgalactic medium (CGM), subtracted as a fixed constant (see §7.3).
- $DM_{\text{IGM}}(z)$ is the dispersion measure accumulated during propagation through the ionized intergalactic medium (IGM).
- $\frac{DM_{\text{host}}}{1+z}$ is the contribution from the host galaxy and its local environment. The factor of $\frac{1}{1+z}$ redshifts the local host-frame dispersion measure to the observer-frame, accounting for the cosmological expansion.
- $DM_X(M_i, b_{\perp})$ is the rest-frame contribution from the i^{th} intersected halo of the N_X intervening halos.

Also, we define

$$DM_{\text{halo}} = \sum_i^{N_X} \frac{DM_X(M_i, b_{\perp})}{1+z_i} \quad (5.2)$$

$$DM_{\text{cosmic}} = DM_{\text{IGM}} + DM_{\text{halo}} \quad (5.3)$$

$$DM_{\text{MW}} = DM_{\text{MW,halo}} + DM_{\text{MW,ISM}} \quad (5.4)$$

$$DM_{\text{ex}} = DM_{\text{cosmic}} + \frac{DM_{\text{host}}}{1+z} = DM_{\text{obs}} - DM_{\text{MW}} \quad (5.5)$$

The intergalactic medium contribution (DM_{IGM}) accumulates through propagation via the ionized cosmic web. Its redshift-dependent expectation value is described by the **Macquart relation**[4]:

$$\langle DM_{\text{IGM}}(z) \rangle = \frac{3cH_0\Omega_b}{8\pi Gm_p} \times \int_0^z \frac{f_d(z') f_{\text{IGM}}(z') (1+z')}{\sqrt{\Omega_m(1+z')^3 + \Omega_\Lambda}} dz' \quad (5.6)$$

Derivation of the Macquart Relation

The cosmological *scale factor* $a(t)$ describes the relative expansion of the Universe at cosmic time t , normalised to unity today: $a_0 \equiv a(t_0) = 1$. The cosmological redshift is defined by:

$$1+z = \frac{a_0}{a(t)} = \frac{1}{a(t)} \quad (5.7)$$

Differentiating both sides with respect to t , and using the Hubble parameter $H(z) \equiv \dot{a}/a$:

$$dz = -\frac{\dot{a}}{a^2} dt = -\frac{H(z)}{a} dt = -(1+z)H(z) dt \Rightarrow dt = -\frac{dz}{(1+z)H(z)} \quad (5.8)$$

For a photon propagating along a null geodesic in the flat FLRW^a metric ($ds^2 = 0$), the proper path length element traversed in proper time $|dt|$ is:

$$dl = c|dt| = \frac{c dz}{(1+z)H(z)} \quad (5.9)$$

Because the Universe expands homogeneously, the comoving baryon number density is conserved. The *physical* baryon number density at redshift z therefore scales as:

$$n_b(z) = n_{b,0} (1+z)^3 \quad (5.10)$$

where $n_{b,0}$ is today's mean baryon number density. Not all baryons contribute free electrons to the IGM: a fraction $f_{\text{IGM}}(z)$ of cosmic baryons reside in the diffuse IGM, and of those, a fraction $f_d(z)$ are ionized. The physical free-electron number density is therefore:

$$n_e(z) = f_{\text{IGM}}(z) f_d(z) n_{b,0} (1+z)^3 \quad (5.11)$$

The Friedmann equation at $z = 0$, for a flat Universe, gives the critical density today:

$$\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G} \quad (5.12)$$

The mean baryon mass density today is $\rho_{b,0} = \Omega_b \rho_{\text{crit},0}$. Approximating the mean baryon mass by the proton mass m_p :

$$n_{b,0} = \frac{\rho_{b,0}}{m_p} = \frac{\Omega_b \rho_{\text{crit},0}}{m_p} = \frac{3H_0^2 \Omega_b}{8\pi G m_p} \quad (5.13)$$

DM is an observer-frame quantity, inferred from $\Delta t_{\text{obs}} \propto v_{\text{obs}}^{-2} DM_{\text{obs}}$. Consider a thin plasma slab at redshift z' with proper depth dl' . In the rest-frame of that slab, the photon frequency is $\nu' = \nu_{\text{obs}}(1+z')$,

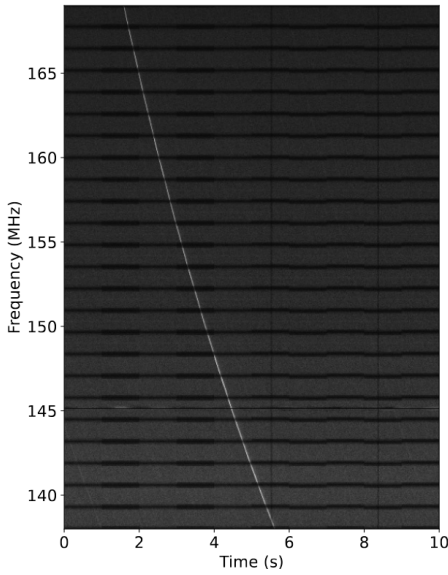


Figure 2: Dynamic spectrum showing the observed signal intensity (of a giant pulse from the Crab pulsar) as a function of frequency and arrival time (Bassa et al. 2017).

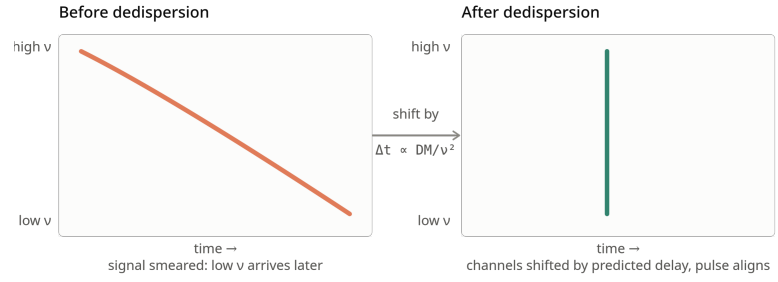


Figure 3: Schematic representation showing dispersed dynamic spectrum before and after dedispersion.

so the local dispersion delay produced by that slab is:

$$\delta t' \propto \frac{n_e(z') dl'}{v'^2} = \frac{n_e(z') dl'}{v_{\text{obs}}^2 (1+z')^2} \quad (5.14)$$

Cosmological time dilation then stretches this delay by $(1+z')$ when received by the observer:

$$\delta t_{\text{obs}} = (1+z') \delta t' \propto \frac{n_e(z') dl'}{v_{\text{obs}}^2 (1+z')^2} = \frac{1}{v_{\text{obs}}^2} \times \frac{n_e(z') dl'}{(1+z')^2} \quad (5.15)$$

We define :

$$dDM_{\text{obs}} = \frac{n_e(z')}{1+z'} dl' \quad (5.16)$$

Substituting the results into $DM_{\text{IGM}} = \int dDM_{\text{obs}}$:

$$\begin{aligned} DM_{\text{IGM}}(z) &= \int_0^z \frac{n_e(z')}{1+z'} \cdot \frac{c}{(1+z')^2 H(z')} dz' \\ &= c n_{b,0} \int_0^z \frac{f_{\text{IGM}}(z') f_d(z') (1+z')^3}{(1+z')^2 H(z')} dz' \\ &= c n_{b,0} \int_0^z \frac{f_{\text{IGM}}(z') f_d(z') (1+z')}{H(z')} dz' \end{aligned} \quad (5.17)$$

Using the flat- Λ CDM Hubble parameter $H(z') = H_0 \sqrt{\Omega_m (1+z')^3 + \Omega_\Lambda}$ and substituting $n_{b,0}$, we obtain the final expression of the Macquart relation:

$$\begin{aligned} \langle DM_{\text{IGM}}(z) \rangle &= \frac{3 c H_0 \Omega_b}{8 \pi G m_p} \\ &\times \int_0^z \frac{f_{\text{IGM}}(z') f_d(z') (1+z')}{\sqrt{\Omega_m (1+z')^3 + \Omega_\Lambda}} dz' \end{aligned} \quad (5.18)$$

^aFriedmann–Lemaître–Robertson–Walker (FLRW) metric

where the cosmological and physical terms are defined as follows:

- c is the speed of light in a vacuum.
- H_0 is the Hubble constant, representing the current expansion rate of the Universe.
- Ω_b is the cosmological baryon density parameter.
- G is the gravitational constant.

- m_p is the proton mass.

- $f_d(z')$ is the free electron fraction of the intergalactic gas, predominantly consisting of Hydrogen and Helium gas. It is defined analytically as [8]:

$$f_d(z) = \frac{3}{4} y_1 \chi_{e,H}(z) + \frac{1}{8} y_2 \chi_{e,He}(z) \quad (5.19)$$

where $y_1 \approx 1$ and $y_2 \approx 1$ are normalization constants for the primordial composition (75% Hydrogen and 25% Helium by mass), and $\chi_{e,H}(z)$, $\chi_{e,He}(z)$ represent the ionization fractions of Hydrogen and Helium. Under the physical assumption of a fully ionized intergalactic medium at low redshifts ($z < 2$), this fraction evaluates to:

$$f_d = \frac{3}{4} (1) + \frac{1}{8} (1) = 0.875 \quad (5.20)$$

If the assumption of complete ionization is violated, the free electron fraction will drop, leading to an underestimation of the inferred cosmic baryon density (Ω_b).

- $f_{\text{IGM}}(z')$ is the fraction of cosmic baryons residing in the diffuse IGM.
- Ω_m and Ω_Λ are the cosmological density parameters for matter and dark energy, respectively. Here, $\Omega_\Lambda = 0.685$ and $\Omega_m = 0.315$.

This suggests that the diffuse intergalactic gas accounts for the primary redshift scaling of dispersion measures, while sightline scatter is dominated by stochastic host galaxy contributions.

Because the local plasma environment of the host galaxy varies, the host contribution (DM_{host}) is modeled as a stochastic variable following a log-normal probability density function[5]:

$$\begin{aligned} P(DM_{\text{host}} | \mu_{\text{host}}, \sigma_{\text{host}}) &= \frac{1}{DM_{\text{host}} \sigma_{\text{host}} \sqrt{2\pi}} \\ &\times \exp \left(-\frac{(\ln DM_{\text{host}} - \mu_{\text{host}})^2}{2\sigma_{\text{host}}^2} \right) \end{aligned} \quad (5.21)$$

where $e^{\mu_{\text{host}}}$ represents the median host dispersion measure, and σ_{host} represents the log-normal dispersion width. A log-normal distribution is appropriate because:

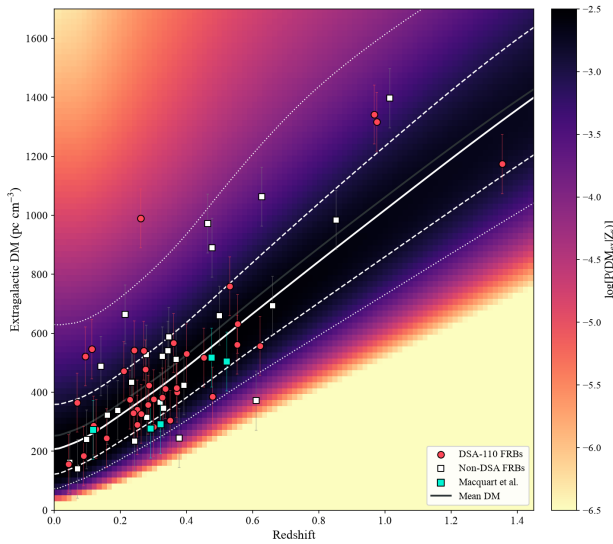


Figure 4: Macquart relation. DSA-110 (red circles), Non-DSA (white squares), and the original Macquart gold sample (cyan squares). The background shading shows the convolved 2D likelihood $\log_{10}[P(DM_{\text{ex}}|z)]$. The solid white line is the median, the dashed white lines are the 1σ contours, the dotted white lines are the 2σ contours, and the solid dark line is the mean $\langle DM_{\text{IGM}} \rangle(z)$ curve. The outlier FRB 20190520B is excluded.

- DM_{host} is positive definite.
- Large values are very rare, so the distribution has a long positive tail.

To infer cosmological and host parameters, the joint likelihood $\mathcal{L}_i$ for each FRB i is calculated by convolving the intergalactic probability density $P(DM_{\text{IGM}}|z)$ (modeled from IllustrisTNG 300-1 simulation sightlines [3]) with the host log-normal distribution:

$$\mathcal{L}_i = P(DM_{\text{obs}}|z_i) = P(DM_{\text{cosmic}} + DM_{\text{host}}|z_i) \quad (5.22)$$

so,

$$\mathcal{L}_i = \int_0^{DM_{\text{obs},i}} P(DM_{\text{cosmic},i} - DM'|z_i) \times P(DM'|\mu_{\text{host}}, \sigma_{\text{host}}) dDM' \quad (5.23)$$

where $DM_{\text{cosmic},i} = DM_{\text{obs}} - DM_{\text{MW,ISM}} - DM_{\text{MW,halo}}$. Thus,

$$\mathcal{L} = \prod_i \int_0^{DM_{\text{cosmic},i}} P(DM_{\text{obs},i} - DM'|z_i) \times P(DM'|\mu_{\text{host}}, \sigma_{\text{host}}) dDM' \quad (5.24)$$

6 Profile of Organisation/Research Lab

This project was conducted within the Astronomy and Astrophysics group led by Prof. Prateek Sharma, in the Department of Physics at the Indian Institute of Science (IISc), Bengaluru. The Department of Physics at Indian Institute of Science (IISc), Bengaluru, established in 1933 under Nobel Laureate Sir C.V. Raman, is a leading centre of physical science research in India, with active programmes spanning condensed matter, high-energy physics, astrophysics, and plasma physics. Within this environment, Prof. Sharma's group actively works on all aspects of the CGM, the diffuse medium around Milky-Way like galaxies that contains the majority of the halo baryons. They have used high-resolution idealized simulations, complementary to the cosmological galaxy formation simulations that generally do not have a high enough resolution to faithfully simulate the multiphase CGM[9].

7 Methodology Followed

7.1 Data Aggregation & Standardization Pipeline

To build a homogeneous dataset, we aggregated localized FRB measurements from the CRAFT [10], MeerTRAP [11], DSA-110 [12], and Connor et al. [5] catalogs, supplemented by public alerts on the Transient Name Server (TNS). Each catalog was standardized and then merged through the following sequential steps:

1. **Schema and column pruning:** Retained only key physical parameters (redshift, DM, coordinates, survey metadata); removed NaN rows in critical fields.
2. **Source ID normalization:** Stripped FRB/FRB prefixes; resolved suffix discrepancies across catalogs (e.g., CRAFT's 20180924B → TNS 20180924A); removed duplicates via redshift and angular-coordinate closeness filters.
3. **Coordinate standardization:** Converted all sexagesimal and Galactic coordinates to decimal degrees (RA, Dec J2000).
4. **Data type normalization:** Mapped heterogeneous repeater flags (Y/N, yes/no, sub-name designations) to a boolean `True/False`; stripped residual LaTeX and parsing artefacts.
5. **Fit-to-burst DM for CHIME:** Used `dm_fitb` (profile-model fit) rather than `bonsai_dm` (coarse online trigger), as the former accounts for sub-burst frequency-dependent structure and yields a higher-fidelity physical measurement [13].¹
6. **Source priority hierarchy:** When a source appeared in multiple catalogs, primary survey publications (CRAFT, MeerTRAP, DSA-110) took precedence over TNS to preserve direct spectroscopic measurements.
7. **MeerTRAP NE2001 back-filling:** Nine MeerTRAP sources [11] were absent from TNS, leaving $DM_{\text{MW,ISM}}$ as NaN; values were manually extracted from the publication and back-filled.
8. **Unique TNS/FRBCAT integration:** Cross-matched against TNS and legacy FRBCAT; identified 2 unique events not already in the primary catalogs (FRB 20230808F and FRB 20210405I) and added them.

7.2 Selection Cuts and Cleaning Filters

To isolate the cosmological dispersion measure and minimize selection effects, the following sequential filters were implemented:

7.2.1 Galactic Foreground Cuts

Two complementary filters are applied to suppress Galactic contamination.

1. All sightlines close to the Galactic disk ($|b| \leq 20^\circ$) are removed to minimize ISM modelling errors from the NE2001 and YMW16 electron density models, which are least reliable at low latitudes.
2. Nearby sightlines where the Milky Way ISM contribution dominates the observed DM, defined as $f_{\text{MW}} = DM_{\text{MW,ISM}}/DM_{\text{obs}} \geq 0.4$, are also excluded (e.g., dropping FRB 20220319D), since the extragalactic signal is too small to meaningfully constrain cosmological parameters.

7.2.2 Signal-to-Noise Ratio Cut

The signal-to-noise ratio (SNR) is a critical diagnostic of detection quality [14]. Low-SNR FRB events indicate noisy, uncertain signals where fitting a mathematical profile model to the burst shape is unreliable and yields high systematic errors in the fit-to-burst dispersion measure (DM_{fitb}).

¹<https://www.chime-frb.ca/catalog>

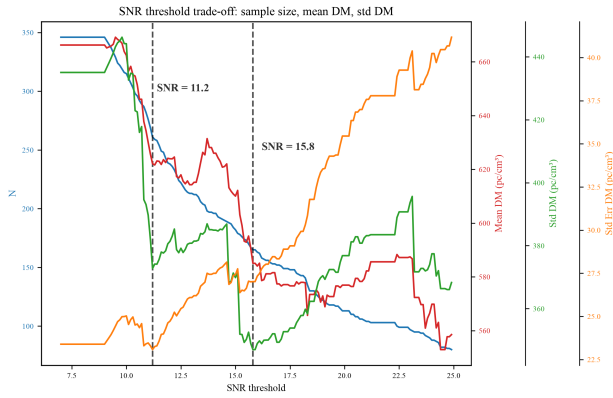


Figure 5: Sensitivity analysis of the FRB sample properties as a function of the signal-to-noise ratio (SNR) threshold. The plot shows the trade-off between the remaining sample size N (blue), mean DM (red), standard deviation of DM (green), and the standard error of DM (orange).

Prior to cleaning, the raw database has a mean dispersion measure of $\langle DM_{\text{obs}} \rangle = 674.77 \text{ pc cm}^{-3}$. Applying the Galactic latitude disk cut and baseline SNR cleaning reduces the mean dispersion measure to $\langle DM_{\text{obs}} \rangle = 656.54 \text{ pc cm}^{-3}$. This decrease is physically expected because high-DM (high-redshift) bursts are intrinsically more distant and undergo more geometric dilution and scattering, making them more likely to be detected with low SNRs.

To determine the optimal SNR threshold, we perform a sensitivity analysis comparing two data-driven candidate cutoffs:

- **SNR = 11.2:** At this cutoff, the standard error of the mean (SE), which measures the statistical precision of our sample mean, achieves its global minimum, maximizing our confidence in the mean relation. The standard deviation (SD) also reaches a local minimum. This threshold retains a large sample size. This represents a rigorous, data-driven choice rather than an arbitrary conventional round number, optimizing the trade-off between sample size and measurement confidence.
- **SNR = 15.8:** Although the standard deviation reaches its lowest point here, the standard error is higher and increasing, and the sample size drops significantly. Crucially, the mean DM decreases significantly at this threshold, reflecting the deletion of a large number of high-DM FRBs. Removing these events biases cosmological inferences toward low-DM (nearby) FRBs and discards the high-redshift population where the Macquart relation's cosmological signal is most diagnostic [4].

Based on these statistics, we adopt $SNR = 11.2$ as our final signal-to-noise ratio threshold, optimizing the statistical precision of our cosmological sample².

An important physical observation is that even after applying the signal-to-noise ratio threshold and Galactic latitude cuts, the overall distribution shape of the dispersion measures remains remarkably identical. As shown in Figure 6, both the pre-cut and post-cut populations exhibit the characteristic right-skewed shape with a well-defined peak around $300 - 500 \text{ pc cm}^{-3}$ and a long, thin tail extending to high dispersion measures, representing the physical rarity of extreme dispersion events. The primary difference is a slight leftward shift of the distribution's peak and mean (from $\langle DM_{\text{obs}} \rangle = 674.77$ to $656.54 \text{ pc cm}^{-3}$), indicating the successful subtraction of the Milky Way disk ISM contribution. This confirms that the extragalactic features dominate the overall dispersion signal, providing a clean probe for cosmological modeling. Having fixed the SNR threshold,

²This test is not a substitute for Jackknife resampling test. It protects the data quality for the Jackknife test to be meaningful which assumes the data points to be individually valid and tests for the robustness.

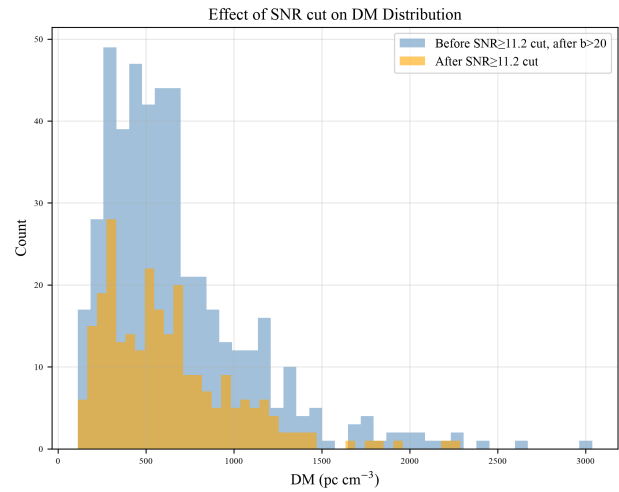


Figure 6: Dispersion measure (DM) distributions of the sample before and after the $SNR \geq 11.2$ selection cut. The overall right-skewed shape and high-DM tail remain identical, while the peak shifts slightly leftward due to Galactic foreground subtraction.

we next address a second selection-bias source: the treatment of repeating FRB sources within the catalog.

7.2.3 Repeater Handling and Selection Bias Mitigation

Initially, complete exclusion of repeating FRBs was considered. Physically, the plasma in the host environments of repeaters can evolve over time, meaning different bursts probe slightly different plasma states, and simple DM averaging would conflate these distinct physical measurements. Crucially, if we do not take care of repeating sources by grouping and averaging their bursts, the sample statistics become heavily biased toward the most active repeating sources (such as FRB20121102A or FRB 20190520B), which have emitted hundreds of detected bursts. Furthermore, repeaters may reside in physically different environments (such as younger, denser, or more highly magnetized circumgalactic structures), and mixing them directly without averaging or careful treatment would severely skew the DM distribution and bias the cosmological inferences. A boxplot comparison of the repeater and non-repeater populations in CHIME Catalog 1 [13]³ shows that repeaters are systematically clustered at lower DMs, with a median of $\sim 340 \text{ pc cm}^{-3}$ compared to $\sim 510 \text{ pc cm}^{-3}$ for non-repeater counterparts (a difference of $\sim 170 \text{ pc cm}^{-3}$). This is physically expected since lower-DM (closer) sources are easier to detect repeatedly. Excluding repeaters entirely would artificially skew the sample toward higher DMs. Thus, repeating sources are retained in the final sample by averaging their bursts, using their mean DMs to mitigate this selection bias. This is justified because the observed DM values for individual bursts of each repeating source are highly consistent with each other, showing very little variation.

7.2.4 Outlier Removal

Identified and removed the anomalous source FRB20190520B ($z = 0.241$, $DM_{\text{ex}} \approx 1116.1 \text{ pc cm}^{-3}$). Its host contribution is estimated at $\approx 900 \text{ pc cm}^{-3}$ (nearly 5 times the expectation), which would heavily skew the host log-normal parameter estimation. The final cleaned catalog contains $N = 80$ eligible localized FRBs spanning a redshift range of $0.0008 \leq z \leq 1.354$.

7.3 Galactic and Halo Modeling

To retrieve the true cosmological and host dispersion signals, we must isolate the extragalactic dispersion measure DM_{ex} from the observed

³In the CHIME Catalog 1 database, repeaters are programmatically marked and flagged with a value of -9999 in the sub-burst and associated columns to distinguish them from single-burst events.

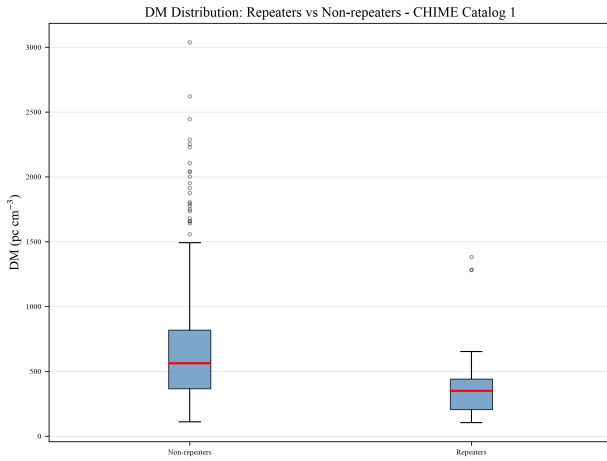


Figure 7: Dispersion measure (DM) distributions for repeaters and non-repeaters in the CHIME/FRB Catalog 1 [13], demonstrating the lower median DM of the repeating population ($\sim 340 \text{ pc cm}^{-3}$ vs. $\sim 510 \text{ pc cm}^{-3}$) and the statistical selection bias introduced if repeaters are excluded.

dispersion DM_{obs} . For modeling the Galactic disk ISM contribution ($DM_{\text{MW,ISM}}$), two primary electron density models are compared:

- **NE2001 [6]:** Built using pulsar dispersion and scattering data, this model has been the standard baseline in cosmological analyses.
- **YMW16 [7]:** A more recent alternative utilizing a larger pulsar dataset and updated spiral arm structure geometries.

A comparison of the excess dispersion measures derived from these two Galactic models shows that the values lie almost exactly along the 1 : 1 line, demonstrating strong agreement for the majority of high-Galactic latitude FRBs in our sample in Figure 9. The mean difference between the models is small:

$$\langle DM_{\text{exc,NE2001}} - DM_{\text{exc,YMW16}} \rangle \approx 4.71 \text{ pc cm}^{-3} \quad (7.1)$$

However, noticeable scatter away from the 1 : 1 line is observed. This modeling systematic occurs primarily near the Galactic plane ($|b| < 20^\circ$) or within dense Galactic regions, where clumpy local plasma features make the electron density profile extremely difficult to model. Furthermore, neither the NE2001 nor the YMW16 model includes the diffuse gas residing in the extended Galactic halo. Manually subtracting the circumgalactic halo contribution is therefore required. While several models of the Galactic halo suggest a gas contribution of $DM_{\text{MW,halo}} \approx 50 \text{ pc cm}^{-3}$, we adopt a standardized value of:

$$DM_{\text{MW,halo}} = 30 \text{ pc cm}^{-3} \quad (7.2)$$

to maintain consistency with the baseline conventions of Connor et al. 2025 [5]. This subtraction is modeled as a constant, isotropic contribution because high-Galactic latitude sightlines exhibit negligible angular variation in Galactic halo models, and the outer halo’s properties are poorly constrained compared to the disk, making a fixed subtraction a robust first-order approximation. Crucially, since $DM_{\text{MW,halo}}$ enters the dispersion measure partitioning as an additive constant, any residual deviation in the true Galactic halo contribution is absorbed directly into the inferred rest-frame host galaxy contribution.

7.4 Convolved Likelihood

Following the DM budget partitioning defined in Equation 5.1 and the Galactic foreground subtraction described in §7.3, assuming that the host galaxy contribution DM_{host} is physically independent of the cosmic pair (DM_{cosmic}), the PDF of their sum is given by the mathematical convolution

of their individual PDFs:

$$\begin{aligned} P(DM_{\text{ex}} | z, \theta) &= (P_{\text{cosmic}} * P_{\text{host}})(DM_{\text{ex}} | z, \theta) \\ &= \int_0^{DM_{\text{ex}}} P_{\text{cosmic}}(DM_{\text{ex}} - DM' | z) \\ &\quad \times P_{\text{host}}(DM' | z, \mu_{\text{host}}, \sigma_{\text{host}}) dDM' \end{aligned} \quad (7.3)$$

The joint cosmic component is modeled as a bivariate, correlated log-normal distribution in the halo and IGM dispersion contributions:

$$\begin{aligned} P_{\text{cosmic}}(DM_{\text{halo}}, DM_{\text{IGM}} | z, \theta) &= \frac{A}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}} \\ &\quad \times \frac{\exp\left(-\frac{w_x^2 - 2\rho w_x w_y + w_y^2}{2(1-\rho^2)}\right)}{DM_{\text{halo}} \cdot DM_{\text{IGM}}} \end{aligned} \quad (7.4)$$

where $w_x = (\ln DM_{\text{halo}} - \mu_x)/\sigma_x$ and $w_y = (\ln DM_{\text{IGM}} - \mu_y)/\sigma_y$, and ρ is the correlation coefficient between the two components.

Computational Implementation: In its raw formulation, calculating the convolved likelihood requires a double numerical integration over the 2D cosmic grid at each MCMC step for every FRB:

$$\begin{aligned} P(DM_{\text{ex}} | z, \theta) &\propto \int_0^{DM_{\text{ex}}} dDM_{\text{IGM}} \int_0^{DM_{\text{ex}} - DM_{\text{IGM}}} \\ &\quad dDM_{\text{halo}} \times P_{\text{cosmic}}(DM_{\text{halo}}, DM_{\text{IGM}} | z) \\ &\quad \times P_{\text{host}}(DM_{\text{ex}} - DM_{\text{halo}} - DM_{\text{IGM}} | z) \end{aligned} \quad (7.5)$$

In the original code, this was evaluated as a $\mathcal{O}(n_{\text{FRB}} n_{\text{DM}}^3)$ 3D multiplication over a ‘(100, 100, 100)’ grid, allocating a 3D intermediate array of 1,000,000 floats 100 times per likelihood call, causing massive memory bandwidth and CPU bottlenecking.

To accelerate the solver, we implemented a discrete grid factorization based on a physical conservation insight: the host galaxy contribution $DM_{\text{host}} = (DM_{\text{ex}} - DM_{\text{cosmic}})(1+z)$ only depends on the combined sum of the cosmic components $DM_{\text{cosmic}} = DM_{\text{halo}} + DM_{\text{IGM}}$, rather than on their individual partitions. Because the grid resolutions for DM_{halo} and DM_{IGM} are identical linear arrays of size $N \times N$, lines of constant DM_{cosmic} are exactly the anti-diagonals of the 2D cosmic PDF grid.

We collapse the 2D grid ‘(100, 100)’ into a 1D array of unique cosmic sums D_s of size ‘(199,)’ by summing the cosmic probabilities along the anti-diagonals first:

$$\begin{aligned} P_{\text{cosmic, sum}}(D_s | z, \theta) &= \sum_{i+j=D_s} P_{\text{cosmic}}(DM_{\text{halo}, j}, \\ &\quad DM_{\text{IGM}, i} | z, \theta) \end{aligned} \quad (7.6)$$

We then construct a pre-computed 2D host galaxy probability matrix $P_{\text{host, 2D}}$ of shape ‘(100, 199)’ (observed extragalactic DM vs unique cosmic sums) and evaluate the convolved probability as a single matrix-vector product:

$$P(DM_{\text{ex}} | z, \theta) = P_{\text{host, 2D}} \cdot P_{\text{cosmic, sum}} \quad (7.7)$$

This anti-diagonal reduction reduces the operation to 100×199 steps, completely avoids $\mathcal{O}(N^3)$ memory allocations, and yields about a 22-fold CPU execution speedup, reducing the end-to-end 2000-step production run from 45 hours to ≈ 2 hours. To prevent log-posterior numerical crashes for low-DM sightlines, the lower grid integration limit is regularized to $DM_{\text{min}} = 1.0 \text{ pc cm}^{-3}$.

7.5 Use of IllustrisTNG Simulations

The IllustrisTNG hydrodynamical simulation (TNG300-1 run) [3] forms the basis of the joint cosmic distribution P_{cosmic} at each redshift z . The TNG300-1 box has a side length of $205 h^{-1} \text{ Mpc}$ comoving and tracks the hydrodynamical evolution of 2500^3 dark matter particles and 2500^3 gas cells.

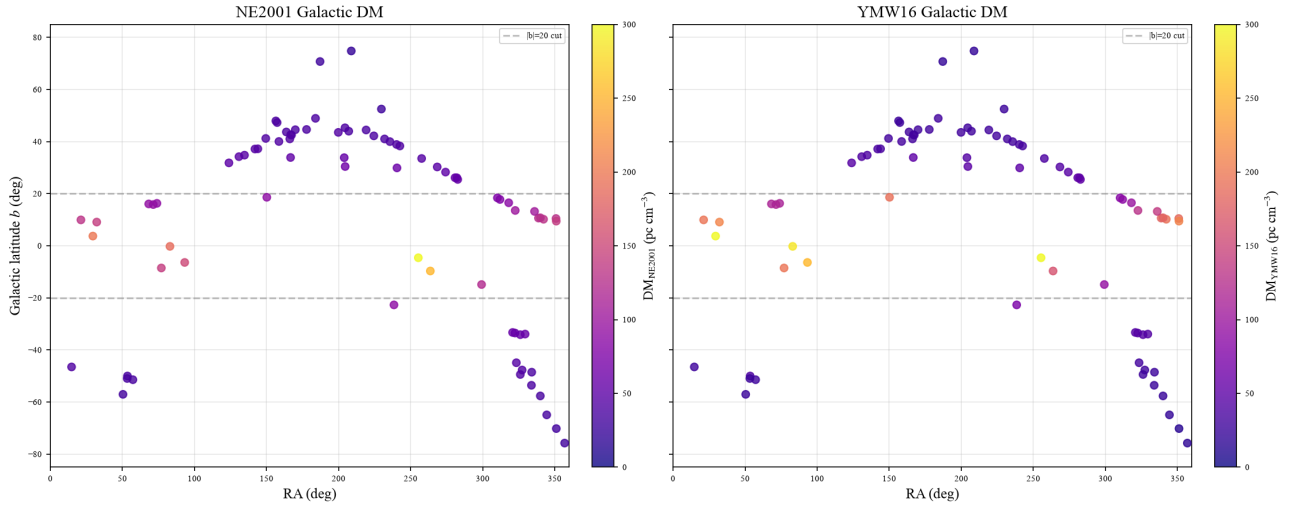


Figure 8: Comparison of the $DM_{MW,ISM}$ derived using the NE2001 and YMW16 Galactic disk ISM models for our localized sample.

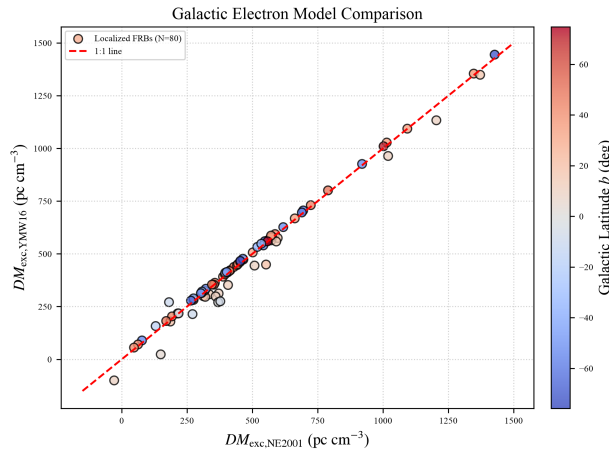


Figure 9: Comparison of the excess dispersion measures derived using the NE2001 and YMW16 Galactic disk ISM models for our localized sample. The dashed red line represents the 1:1 line of perfect agreement.

Thousands of mock sightlines are integrated through the simulation box at redshift snapshots ($z = 0.1, 0.2, \dots, 5.0$). For each sightline, the electron density is partitioned into:

- **Diffuse IGM (DM_{IGM}):** Intergalactic plasma outside massive halos ($M_{\text{halo}} < 10^{12} M_{\odot}$ or outside R_{500}).
- **Collapsed Halos (DM_{halo}):** Ionized gas within the CGM of intervening halos ($M_{\text{halo}} \geq 10^{12} M_{\odot}$ or within R_{500}).

By fitting the joint $(DM_{IGM}, DM_{\text{halo}})$ distributions at each snapshot, the 6 bivariate log-normal parameters $(\mu_x, \mu_y, \sigma_x, \sigma_y, \rho, A)$ are tabulated as a function of redshift and interpolated to each FRB's redshift z_i using 1D cubic splines during MCMC sampling.

To avoid re-running new simulations for every trial value of $(f_{IGM}, f_{\text{halo}})$, we rescale the log-means analytically relative to the baseline TNG values ($f_{IGM, \text{TNG}} = 0.827, f_{\text{halo}, \text{TNG}} = 0.138$) [5]⁴:

$$\mu_y(z) = \mu_{y, \text{TNG}}(z) + \ln\left(\frac{f_{IGM}}{f_{IGM, \text{TNG}}}\right) \quad (7.8)$$

$$\mu_x(z) = \mu_{x, \text{TNG}}(z) + \ln\left(\frac{f_{\text{halo}}}{f_{\text{halo}, \text{TNG}}}\right) \quad (7.9)$$

⁴This assumes a first-order approximation: the shape of the distribution (ρ, σ) and the covariance structure of density fluctuations remain fixed; only the normalization (log-mean) changes.

Physically, it just rescales the typical amount of gas in the IGM channel keeping the distribution invariant.

7.6 Alternative Host Galaxy Modeling: The Gamma Distribution

While the log-normal distribution is the standard choice in the primary literature [4, 5], we also implement and test a Gamma distribution model for the host galaxy DM.

Physically, the Gamma distribution is motivated by viewing the host interstellar medium (ISM) and circumgalactic medium (CGM) not as a smooth continuous fluid, but as a collection of discrete ionized clouds. If a line of sight through a host galaxy intersects a random, discrete number of ionized clouds, and each cloud contributes an independent and exponentially distributed dispersion measure.

$$\delta DM_i \sim \text{Exp}(\theta) \quad (7.10)$$

$$\text{Exp}(\theta) = \begin{cases} \theta e^{-\theta x}, & x \geq 0, \\ 0, & x < 0. \end{cases} \quad (7.11)$$

Then the total host contribution is the sum of these contributions:

$$DM_{\text{host}} = \sum_{i=1}^k \delta DM_i \sim \Gamma(k, \theta) \quad (7.12)$$

Mathematically, the sum of k independent and identically distributed exponential random variables follows an Erlang distribution⁵. Generalizing to non-integer values of the shape parameter k (which can phenomenologically represent geometric projection angles ϕ , clump size distributions, and partial line-of-sight edge clipping), the host galaxy DM in its rest-frame follows a Gamma distribution:

$$P_{\text{host}}(DM_{\text{host}} | k, \theta) = \frac{1}{\theta^k \Gamma(k)} (DM_{\text{host}})^{k-1} \times e^{-DM_{\text{host}}/\theta} \quad (7.13)$$

where $\Gamma(k)$ is the standard Gamma function.

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t} dt \quad (7.14)$$

Here, k is the shape parameter, and θ is the scale parameter, yielding a mean rest-frame host DM of $\langle DM_{\text{host}} \rangle = k\theta$.

This parameterisation introduces several physical and statistical advantages.

⁵Erlang is a special case of Gamma distribution restricted to integer shape parameter i.e., $E(k, \lambda) = \Gamma(k \in \mathbb{Z}^+, \lambda)$.

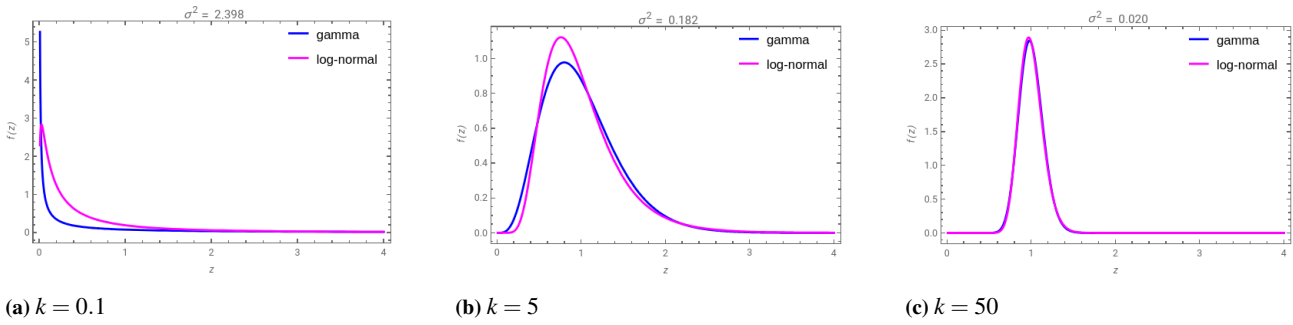


Figure 10: Comparison of the Gamma distribution $\Gamma(k, \theta = \frac{1}{k})$ and Log-normal distribution $L(\mu = 1, \sigma = \ln(1 + \frac{1}{k}))$.

tages:

- **Flexibility at the Origin** ($DM_{\text{host}} \rightarrow 0$): A log-normal PDF is strictly pinned to zero at the origin ($P(DM_{\text{host}}) \rightarrow 0$ rapidly as $DM_{\text{host}} \rightarrow 0$). However, if an FRB is born in the clean outskirts of a host, in a globular cluster, or within a gas-poor dwarf galaxy, its host contribution can be negligible. A Gamma distribution with shape parameter $k \leq 1$ does not drop to zero at the origin; it can be finite or diverge, providing a natural mathematical model for clean, zero-host sightlines.
- **Lighter Tails and Outlier Mitigation:** The Gamma distribution exhibits an exponentially decaying tail (scaling as $e^{-DM_{\text{host}}/\theta}$), which is lighter than the sub-exponential, heavy tail of the log-normal distribution. This makes parameter estimation less sensitive to extreme local outliers (like FRB20190520B) which otherwise force a log-normal model to inflate its variance, biasing the entire population.

Furthermore, we account for observational selection effects in our priors. Real-world radio telescopes preferentially detect sharp, unscattered pulses, meaning our catalog is biased against FRBs passing through dense galactic disks (large k , high scattering) and biased toward clean sightlines (small k , low scattering). If we assume a flat uniform prior on k , the MCMC sampler will over-represent the heavily scattered, high- k population, leading to an overestimate of the average host DM. Because of the host-cosmic degeneracy, overestimating the host DM forces the MCMC to compensate by underestimating the cosmic slope, artificially depressing f_{IGM} . To correct for this, we implement a selection-biased decaying prior on the shape parameter:

$$p(k) \propto e^{-k/\lambda} \quad (7.15)$$

with decay scale $\lambda = 1.5^6$, which penalizes high- k configurations to reflect the telescope's selection function.

7.7 MCMC Parameter Sampling and Prior Distributions

Using the convolved probability $P(DM_{\text{ex}}|z, \theta)$, the joint likelihood for a catalog of N localized FRBs is:

$$\mathcal{L}(\theta) = \prod_{i=1}^N P(DM_{\text{ex},i}|z_i, \theta) \quad (7.16)$$

We sample the joint posterior distribution $p(\theta|\text{data}) \propto \mathcal{L}(\theta)p(\theta)$ using the affine-invariant ensemble MCMC sampler (`emcee`). The 4-parameter model fits the parameters $\theta = \{f_{\text{IGM}}, f_{\text{halo}}, \mu_{\text{host}}, \sigma_{\text{host}}\}$.

⁶The decay scale $\lambda = 1.5$ is chosen to center the prior expectation of the shape parameter k at 1.5, balancing the physical expectation that detectable, low-scattering sightlines intersect on average 1-2 ionized clumps in the host ISM, while preventing the sampler from entering the unphysical $k < 1$ regime where the Gamma probability density diverges at the origin, and penalizing high- k configurations that correspond to heavily scattered lines of sight.

The priors $p(\theta)$ are defined as[5]:

- Flat prior on $f_{\text{IGM}} \in [0, 1]$.
- Flat prior on $f_{\text{halo}} \in [0, 1]$.
- Flat prior on $\mu_{\text{host}} \in [0, 7.0]$.
- Flat prior on $\sigma_{\text{host}} \in [0.1, 2.5]$.
- A cosmological prior constraint: $f_{\text{IGM}} + f_{\text{halo}} < 1$, representing the physical upper limit of the diffuse cosmic baryon fraction.

The MCMC is initialized with 32 walkers. The sampler is run for 28000 iterations, discarding the first 1500 steps as burn-in. Walker convergence is verified by checking the autocorrelation time (τ), ensuring the production chains are run for multiple tens of τ to guarantee independent, uncorrelated posterior samples.

7.8 Galaxy Overdensity Cross-Correlation: Physical Motivation and Pipeline

When an FRB is observed with an exceptionally high dispersion measure (DM), a fundamental question arises: is a high DM more likely driven by a high cosmological redshift (z) or an intervening galaxy overdensity (δ_{gal})? At very high redshifts, the luminosity distance $d_L(z)$ is very large, meaning only the most intrinsically energetic, rare FRBs can exceed the survey detection threshold. Furthermore, even if such ultra-distant bursts are detected, their SNR is very unlikely to exceed stringent survey cutoffs due to pulse broadening and distance attenuation. Consequently, observed high- DM bursts are statistically far more likely to be caused by an extreme galaxy overdensity along a moderate-redshift sightline rather than an ultra-high redshift source [15]. This underscores why dispersion measure residuals (δ_{DM}) serve as a sensitive, direct probe of localized overdensities. The standard Macquart relation represents a statistical average over a homogeneous Universe. In the real Universe, matter is arranged in a cosmic web of nodes, filaments, and voids. An individual FRB sightline acts as a pencil beam traversing this structured medium:

- **Overdense sightlines:** Intersecting massive clusters or dense filaments, the sightline accumulates excess free electrons, yielding a positive dispersion residual: $\delta_{DM} = DM_{\text{cosmic}} - DM_{\text{Macquart}}(z) > 0$.
- **Underdense sightlines:** Paths through cosmic voids bypass the bulk of the cosmic gas, yielding $\delta_{DM} < 0$.

Galaxies trace the underlying dark matter and baryonic gas distribution, so the cumulative foreground galaxy overdensity $\Sigma\delta_{\text{gal}}$ serves as a normalized proxy for the total gas density along the sightline. Cross-correlating $\Sigma\delta_{\text{gal}}$ with δ_{DM} allows us to observationally verify that the scatter around the Macquart relation is driven by the cosmic web, providing a direct map of the Universe's diffuse baryons.

SDSS Cross-Correlation Pipeline. To search for direct observational evidence that FRBs trace the diffuse cosmological gas, we implemented

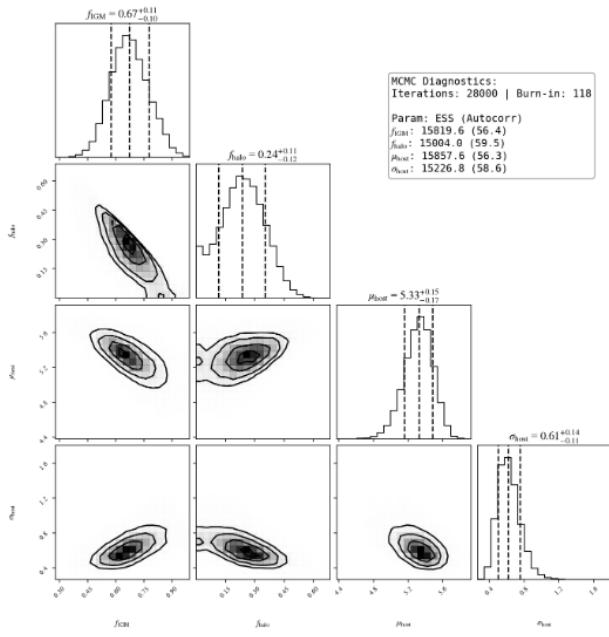


Figure 11: MCMC posterior corner plot showing the joint parameter constraints and correlations for the 4-parameter physical model (IGM fraction f_{IGM} , halo fraction f_{halo} , host galaxy median dispersion $e^{\mu_{\text{host}}}$, and log-normal width σ_{host}) evaluated on the extended catalog ($N = 80$).



Figure 13: Optical images of representative localized FRB sightlines with foreground galaxies retrieved from NED. The red cross denotes the FRB position, the dashed circle marks a projected physical radius of 2 Mpc, and cyan circles indicate foreground galaxies with $z < z_{\text{FRB}}$. The listed δDM values correspond to the FRB excess dispersion measures. This figure is intended to provide physical intuition for the connection between DM residuals and foreground galaxy overdensities; it is not based on the galaxy sample used in the statistical analysis. Owing to the incompleteness of NED, only relatively bright, previously catalogued galaxies are shown, and the true galaxy environment is significantly richer than depicted.

a cross-correlation pipeline between FRB dispersion measures and foreground galaxy overdensities using Sloan Digital Sky Survey (SDSS) data. First, we mapped the sky footprint of the localized FRB sample against the SDSS coverage map, identifying $N_{\text{overlap}} = 18$ localized sources whose sightlines lie within the SDSS footprint (including 7 from DSA-110, 6 from ASKAP, 3 from CHIME, 1 from MeerTRAP, and 1 from Parkes). For each overlapping source, we queried the SDSS database (Data Release 18) for all PhotoObj/Photoz galaxy matches within a $0.3^\circ \times 0.3^\circ$ box centered on the FRB coordinates. To ensure the integrity of the intervening matter mapping, we cleaned the raw galaxy catalogs by applying a photo-z quality cut: we excluded galaxies with invalid or negative photo-z estimations ($z_{\text{photo}} \leq 0$), and discarded sources with a relative photo-z uncertainty

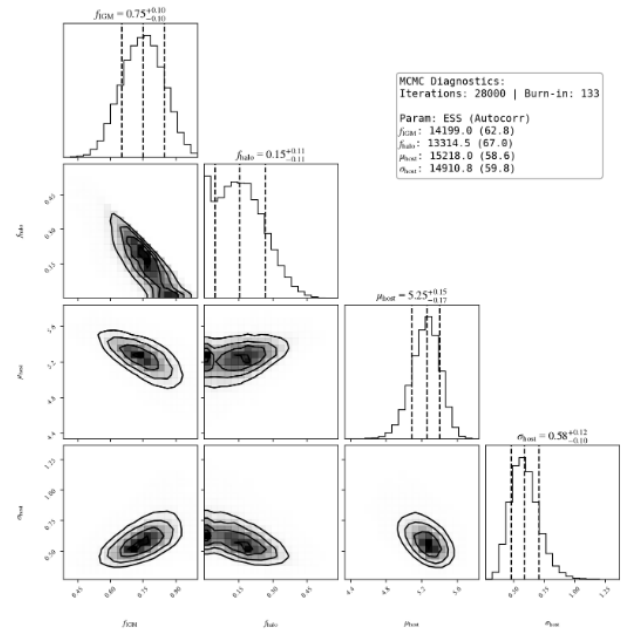


Figure 12: MCMC posterior corner plot showing the joint parameter constraints and correlations for the 4-parameter physical model (IGM fraction f_{IGM} , halo fraction f_{halo} , host galaxy median dispersion $e^{\mu_{\text{host}}}$, and log-normal width σ_{host}) evaluated on the Connor catalog ($N = 69$) for comparison purposes.

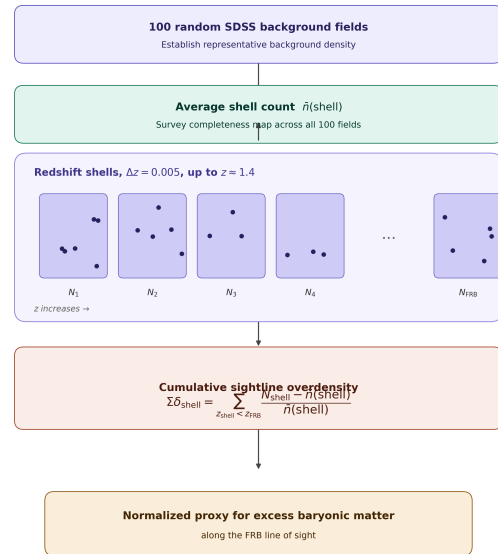


Figure 14: Flow diagram documenting the method for cross-correlation between dispersion measure residual and galaxy overdensity

exceeding:

$$\frac{z_{\text{photo, err}}}{z_{\text{photo}}} \geq 0.10 \quad (7.17)$$

Furthermore, we only retained galaxies satisfying $z_{\text{photo}} < z_{\text{FRB}}$ to isolate the foreground structures along the sightline. A simple total count of foreground galaxies along a sightline is heavily biased: more distant (high-redshift) FRBs naturally have longer paths and cross more galaxies (because of volume effect), and the sensitivity of the SDSS catalog drops sharply at higher redshifts due to the survey flux limit (selection function). To normalize for these effects, we implemented a shell-based cumulative

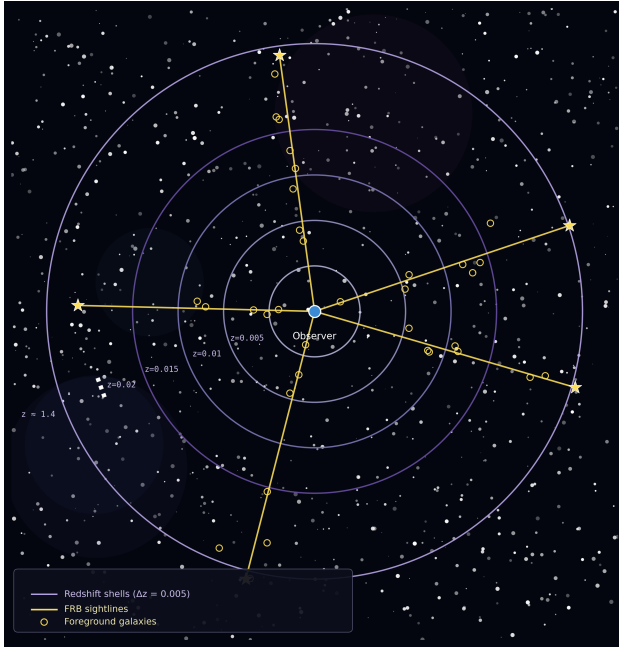


Figure 15: Schematic diagram to explain the redshift binning

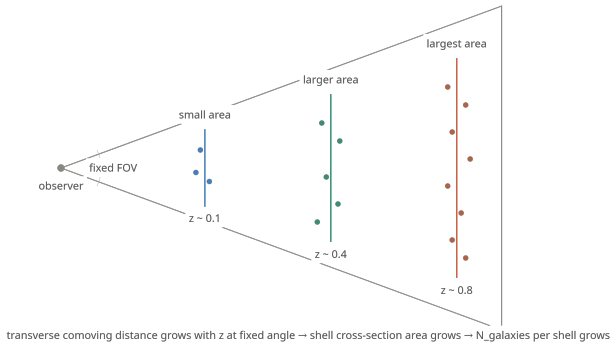


Figure 16: Schematic of survey cone geometry: a fixed angular aperture subtends a larger physical transverse area at higher redshift, increasing the shell volume and raw galaxy counts, which necessitates a shell-by-shell normalized overdensity calculation.

overdensity method:

1. We query and clean galaxy catalogs for 100 random background fields selected within the same SDSS footprint coordinates to establish a representative background density.
2. We divide the redshift space up to $z \approx 1.4$ into discrete shells of width $\Delta z = 0.05$ ⁷.
3. We compute the average expected background galaxy count in each shell, $\bar{n}(\text{shell})$, across all 100 random fields, mapping the survey completeness function.
4. For each actual FRB sightline, the cumulative sightline overdensity $\Sigma \delta_{\text{shell}}$ is calculated by summing the shell-by-shell fractional

⁷A redshift bin width of $\Delta z = 0.05$ is optimal for SDSS photo- z catalogs: it is slightly larger than the typical photo- z uncertainty ($\sigma_z \approx 0.02 - 0.03$), preventing galaxies from randomly scattering across multiple bins; it ensures the average expected count in the background field $\bar{n}$ remains statistically stable to avoid division-by-zero or excessive shot noise in the fractional overdensity $\delta_{\text{shell}} = (N_{\text{shell}} - \bar{n})/\bar{n}$; and it matches the physical scale of cosmic web structures ($\sim 140 h^{-1}$ Mpc radial thickness at $z \approx 0.3$) without averaging out voids and filaments.

overdensities up to the FRB redshift:

$$\Sigma \delta_{\text{shell}} = \sum_{z_{\text{shell}} < z_{\text{FRB}}} \frac{N_{\text{shell}} - \bar{n}(\text{shell})}{\bar{n}(\text{shell})} \quad (7.18)$$

where N_{shell} is the count of foreground galaxies along the FRB sightline in that specific redshift bin. This cumulative overdensity serves as a normalized proxy for the total excess baryonic matter along the path.

DESI Legacy Survey Cross-Correlation Pipeline. To complement the SDSS result and verify signal robustness over a larger footprint with deeper photometry, we extended the same cross-correlation methodology to the DESI Legacy Imaging Survey (Data Release 9). The overall pipeline photo- z quality cuts ($z_{\text{phot, err}}/z_{\text{phot}} < 0.10$), foreground-only galaxy retention ($z_{\text{gal}} < z_{\text{FRB}}$), and shell-based cumulative overdensity calculation $\Sigma \delta_{\text{shell}}$, follows the SDSS procedure described above. The key differences are:

- **Larger overlap sample:** $N_{\text{overlap}} = 39$ sightlines (vs. 18 for SDSS), reflecting DESI's wider sky coverage.
- **Redshift source:** Spectroscopic z_{spec} from DESI/SDSS cross-matches where available; otherwise the random-forest photometric redshift median ($z_{\text{phot, median}}$) from the PRLS value-added catalog [16].
- **Local background calibration:** Rather than 100 global random fields, we generated 3 local background fields per sightline by perturbing the FRB coordinates by $1.2^\circ - 2.2^\circ$, yielding 117 fields total. This accounts for DESI's spatially varying depth more robustly.
- **Queries:** Performed via the Astro Data Lab TAP service.⁸

8 Analysis and Interpretation of Research Project

8.1 IGM Fraction and Host DM Parameter Constraints

We executed the convolved MCMC on both the baseline catalog⁹ ($N = 69$) and the extended Master Catalog ($N = 80$)¹⁰. The posterior constraints are presented in Table 3.

The extended catalog yields $f_{\text{IGM}} = 0.67^{+0.11}_{-0.10}$, showing a decrease of ~ 0.08 compared to the baseline ($0.75^{+0.10}_{-0.10}$). This shift is interpreted as a physical selection effect: the added MeerTRAP sample contains sources with systematically higher observed dispersion measures. To fit these high-DM sightlines, the convolved joint MCMC increases the preferred host galaxy contribution ($\mu_{\text{host}} \approx 5.33$ vs. 5.25 for the baseline and $\sigma_{\text{host}} \approx 0.61$ vs 0.58 for the baseline). By conservation of total dispersion, allocating more DM to the host galaxy leaves a smaller residual portion for the intergalactic medium, pulling down the inferred IGM baryon fraction.

8.2 Instrument-Specific Selection Biases

We ran independent MCMCs for separate telescope instruments to investigate systematic differences. The preferred host galaxy and intergalactic fractions show significant tension¹¹:

- **MeerTRAP Subsample** ($N = 11$): $\mu_{\text{host}} = 6.14^{+0.44}_{-0.64}$, $f_{\text{IGM}} = 0.40^{+0.22}_{-0.23}$.
- **DSA-110 Subsample** ($N = 39$): $\mu_{\text{host}} = 5.35^{+0.18}_{-0.21}$, $f_{\text{IGM}} = 0.70^{+0.12}_{-0.12}$.
- **ASKAP Subsample** ($N = 19$): $\mu_{\text{host}} = 5.35^{+0.36}_{-0.46}$, $f_{\text{IGM}} = 0.70^{+0.15}_{-0.20}$.
- **CHIME Subsample** ($N = 5$): $\mu_{\text{host}} = 5.08^{+0.64}_{-1.91}$, $f_{\text{IGM}} = 0.41^{+0.26}_{-0.26}$.

⁸<https://datalab.noirlab.edu/tap>

⁹Baseline catalog refers to Connor catalog [5]

¹⁰Extended Master Catalog refers to the merged catalog from different sources[5, 10, 11, 12].

¹¹There is also 1 FRB from DSA-10 and 2 FRBs from MeerKAT.

Table 1: Localized FRBs overlapping the SDSS footprint with coordinates, galaxy query counts, DM residuals, and redshifts.

S.No.	FRB-id	RA (HMS)	Dec (DMS)	Query Galaxy Count	δ_{DM} (pc cm ⁻³)	Redshift
1	20180301A	06:12:54.432	04:40:15.960	4383	84.4268	0.3304
2	20180916B	01:58:00.744	65:43:00.480	3104	94.1384	0.0337
3	20190608B	22:16:04.776	-07:53:53.880	11154	183.0734	0.1178
4	20200120E	09:57:54.672	68:49:08.040	3964	16.5066	0.0008
5	20200430A	15:18:49.536	12:22:36.480	18034	199.0532	0.1610
6	20200906A	03:33:59.088	-14:04:59.520	6319	218.9156	0.3688
7	20201124A	05:08:03.504	26:03:38.520	9200	166.1095	0.0982
8	20210807D	19:56:53.136	-00:45:44.640	4650	1.1670	0.1293
9	20211212A	10:29:24.168	01:21:37.800	12576	86.8476	0.0707
10	20220105A	13:55:13.008	22:27:57.960	16650	309.4945	0.2785
11	20220307B	23:23:29.880	72:11:32.640	1315	145.0672	0.2507
12	20220319D	02:08:42.696	71:02:07.080	2010	1.7449	0.0111
13	20220506D	21:12:10.752	72:49:38.280	159	45.7585	0.3005
14	20221219A	17:10:31.152	71:37:36.480	4870	183.9462	0.5540
15	20230216A	10:25:53.328	03:26:12.120	15219	330.5350	0.5310
16	20230307A	11:51:07.584	71:41:42.360	4844	328.8247	0.2710
17	20230521B	23:24:08.640	71:08:16.800	573	63.3090	1.3540
18	20230907D	12:28:34.200	08:39:29.130	15567	599.6654	0.4638

These variations reflect differences in survey frequencies, beam sizes, and host association algorithms. Assuming a single homogeneous log-normal host population across different instruments is physically unrealistic. Combining these populations artificially widens the joint posterior distributions.

8.3 Cosmological Parameter Estimation: $\Omega_b h_{70}$

To place our findings within the broader cosmological context, we employed two distinct statistical methodologies to estimate the cosmological baryon density parameter, $\Omega_b h_{70}$ (or equivalently, Ω_b scaled to a Hubble constant of $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$):

8.3.1 Fixed-Prior Host Model with Analytical $\langle DM_{\text{cosmic}} \rangle$ Likelihood

In this approach, we assume a Gaussian likelihood^[5] over the average cosmic dispersion measure, $\langle DM_{\text{cosmic}} \rangle$, mapping the baryonic slope directly to $\Omega_b h_{70}$ while keeping the host galaxy and IGM scaling parameters fixed at their best-fit reference values. We partition the sample into $K = 5$ quantile redshift bins¹² to construct the likelihood:

It yields $\Omega_b h_{70} = 0.0504^{+0.0048}_{-0.0048}$ on the baseline catalog ($N = 69$), and $\Omega_b h_{70} = 0.0505^{+0.0046}_{-0.0046}$ on the master catalog ($N = 80$). This represents a **4% reduction in the uncertainty interval width**, demonstrating the direct statistical benefit of catalog expansion.

8.3.2 The Impact of the Cosmic Halo Term

During the implementation of the method, we conducted a systematic sensitivity analysis by varying the assumed constant halo term $DM_{\text{MW, halo}}$ to assess its impact on the inferred baryon density and goodness-of-fit:

- **Weak Cosmic Halo** ($DM_{\text{MW, halo}} = 10 \text{ pc cm}^{-3}$): Underestimating the halo gas content only partially flattens the systematic tilt in

¹²The choice of $K = 5$ is set by the trade-off between redshift resolution and statistical validity of the Gaussian likelihood. With our active catalog size ($N = 69 - 80$), $K = 5$ yields approximately 13–16 sources per bin. This is sufficient for the Central Limit Theorem to suppress individual host DM fluctuations, justifying the Gaussian approximation for the sample mean $\langle DM_{\text{cosmic}} \rangle$. Increasing to $K = 10$ would reduce bin occupancy to ~ 7 sources, violating normality and making the binned means highly sensitive to individual outlier sightlines. Decreasing to $K \leq 3$ would remove the redshift resolution needed to detect systematic trends, such as the residual tilt uncovered when the cosmic halo term is omitted. Equal-count quantile binning is used rather than equal-width bins to guard against sparsity in the high-redshift regime.

the residuals, resulting in a fit quality of $\Omega_b h_{70} = 0.0536 \pm 0.0046$ with $\chi^2/\text{ndof} = 3.25/4 = 0.81$.

- **Baseline Cosmic Halo** ($DM_{\text{MW, halo}} = 30 \text{ pc cm}^{-3}$): Introducing this physical constant representing the warm-hot circumgalactic gas phase of intervening halos successfully resolves the systematic tilt. The goodness-of-fit of $\chi^2/\text{ndof} = 3.40/4 = 0.85$, and the inferred value shifts to $\Omega_b h_{70} = 0.0505 \pm 0.0046$, aligning closely with the CMB cosmological fiducial value ($\Omega_b h_{70} \approx 0.047$).
- **Strong Cosmic Halo** ($DM_{\text{MW, halo}} = 50 \text{ pc cm}^{-3}$): Utilizing this upper-limit value suggested by some Milky Way models yields $\Omega_b h_{70} = 0.0473 \pm 0.0046$ and a $\chi^2/\text{ndof} = 3.83/4 = 0.96$.
- **Unconstrained Halo** ($DM_{\text{MW, halo}}$ **Free**): If both parameters are allowed to float freely in a joint 2-parameter fit, it under-predicts the halo contribution ($DM_{\text{MW, halo}} = 9.4 \text{ pc cm}^{-3}$) and forces the baryon slope up to $\Omega_b h_{70} = 0.0537$ ($\chi^2/\text{ndof} = 3.25/3 = 1.08$). This occurs because the simplified algebraic $\langle DM_{\text{cosmic}} \rangle$ binning fails to account for the skewed tail of the log-normal host galaxy distribution, causing the minimizer to attribute the host's excess variance into the constant halo offset.

This quantitatively demonstrates the necessity of the joint MCMC approach.

8.3.3 5-Parameter Joint MCMC

In the first method, the cosmic baryon density ($\Omega_b h_{70}$) is kept fixed to the Planck CMB fiducial value, assuming it is perfectly known. This decouples the cosmological fraction fitting from the total baryon scale and ignores the statistical covariances between $\Omega_b h_{70}$ and the host galaxy profile parameters ($\mu_{\text{host}}, \sigma_{\text{host}}$). To resolve this, we transitioned to a 5-parameter joint MCMC framework, treating $\Omega_b h_{70}$ as a free parameter alongside the fractions and host parameters ($\theta = \{f_{\text{IGM}}, f_{\text{halo}}, \mu_{\text{host}}, \sigma_{\text{host}}, \Omega_b h_{70}\}$). This enables the sampler to fully explore the multi-dimensional parameter space, propagate all joint uncertainties, and marginalize over host galaxy profile degeneracies in a single, self-consistent step, providing a direct, data-driven constraint on the baryon density from the FRB population itself.

It yields $\Omega_b h_{70} = 0.0747^{+0.0397}_{-0.0220}$ on the baseline catalog ($N = 69$), and $\Omega_b h_{70} = 0.0717^{+0.0431}_{-0.0209}$ on the expanded master catalog ($N = 80$). Adding the MeerTRAP and DSA-110 sources in the master catalog results in an

Table 2: Localized FRBs overlapping the DESI Legacy Survey footprint with coordinates, galaxy query counts, DM residuals, and redshifts.

S.No.	FRB-id	RA (HMS)	Dec (DMS)	Query Galaxy Count	δ_{DM} (pc cm ⁻³)	Redshift
1	20180924A	21:44:25.248	-40:54:00.000	30208	38.0195	0.3212
2	20181030A	10:34:23.112	73:45:54.000	3736	29.5050	0.0039
3	20181112A	21:49:23.640	-52:58:15.240	20446	135.2322	0.4755
4	20190523A	13:48:15.600	72:28:10.920	4141	155.7698	0.6600
5	20190608B	22:16:04.776	-07:53:53.880	23695	183.0734	0.1178
6	20191001A	21:33:24.312	-54:44:51.720	26542	251.3181	0.2340
7	20200120E	09:57:54.672	68:49:08.040	6833	16.5066	0.0008
8	20200430A	15:18:49.536	12:22:36.480	20625	199.0532	0.1610
9	20200906A	03:33:59.088	-14:04:59.520	35064	218.9156	0.3688
10	20211212A	10:29:24.168	01:21:37.800	19612	86.8476	0.0707
11	20220105A	13:55:13.008	22:27:57.960	22228	309.4945	0.2785
12	20220204A	18:16:54.312	69:43:21.000	5900	212.4433	0.4000
13	20220310F	08:58:52.920	73:29:26.880	3945	0.6920	0.4790
14	20220330D	10:55:00.288	70:21:02.520	5038	104.2552	0.3714
15	20220418A	14:36:25.344	70:05:45.240	4473	51.0550	0.6220
16	20220509G	18:50:40.800	70:14:37.680	6286	116.1487	0.0894
17	20220914A	18:48:13.632	73:20:12.840	4894	459.6971	0.1138
18	20220920A	16:01:01.704	70:55:07.680	4362	123.2355	0.1585
19	20221012A	18:43:11.688	70:31:27.120	5621	134.6929	0.2840
20	20221027A	08:43:29.280	72:06:03.240	4236	196.9645	0.2290
21	20221029A	09:27:51.240	72:27:09.360	5274	515.0193	0.9750
22	20221219A	17:10:31.152	71:37:36.480	4597	183.9462	0.5540
23	20230124A	15:27:40.080	70:58:05.160	5034	450.7897	0.0940
24	20230216A	10:25:53.328	03:26:12.120	19741	330.5350	0.5310
25	20230307A	11:51:07.584	71:41:42.360	4543	328.8247	0.2710
26	20230626A	15:42:31.104	71:08:00.600	4952	123.5342	0.3270
27	20230628A	11:07:08.808	72:16:54.480	4301	179.3633	0.1265
28	20230712A	11:09:26.040	72:33:28.020	3914	154.8342	0.4525
29	20230808F	03:33:12.960	-51:56:07.019	26314	313.0422	0.3472
30	20230907D	12:28:34.200	08:39:29.130	19191	599.6654	0.4638
31	20231020B	03:49:06.770	-37:46:11.560	29945	505.8922	0.4775
32	20231120A	09:35:56.160	73:17:04.920	4114	312.2844	0.0700
33	20231123B	16:10:09.168	70:47:06.360	4921	120.9604	0.2625
34	20231210F	03:21:37.280	-35:45:41.130	28079	256.5965	0.5000
35	20231220A	08:15:38.088	73:39:35.640	4053	145.8127	0.3355
36	20240119A	14:57:52.128	71:36:42.480	4377	121.1186	0.3700
37	20240213A	11:04:40.392	74:04:31.440	3152	196.9262	0.1185
38	20240215A	17:53:45.912	70:13:56.640	5820	308.4939	0.2100
39	20240229A	11:19:56.040	70:40:34.320	4505	197.6354	0.2870

Table 3: MCMC parameter constraints (median and 68% confidence intervals) comparing the baseline catalog and the master catalog.

Parameter	Baseline Catalog ($N = 69$)	Extended Catalog ($N = 80$)
f_{IGM}	$0.75^{+0.10}_{-0.10}$	$0.67^{+0.11}_{-0.10}$
f_{halo}	$0.15^{+0.11}_{-0.11}$	$0.24^{+0.11}_{-0.12}$
μ_{host}	$5.25^{+0.15}_{-0.17}$	$5.33^{+0.15}_{-0.17}$
σ_{host}	$0.58^{+0.12}_{-0.10}$	$0.61^{+0.14}_{-0.11}$

8.1% widening of the cosmological posterior width. Physically, this widening is driven by degeneracies that arise when the model attempts to fit systematically different host populations jointly (e.g., MeerTRAP favoring a massive host contribution of ~ 421 pc cm⁻³ versus ASKAP’s moderate contribution of ~ 150 pc cm⁻³). This shows that joint unconstrained fits are sensitive to catalog heterogeneity, highlighting the importance of instrument-specific host modeling.

8.4 Host DM Standard Deviation Sensitivity Analysis

Given the structural degeneracies identified in the joint MCMC fits, we conducted a systematic sensitivity analysis to test the stability of the cosmological IGM fraction (f_{IGM}) to the assumed host galaxy dispersion width (σ_{host}). We ran a sequence of 3-parameter MCMCs on the master catalog, fixing $\sigma_{\text{host}} \in \{0.0, 0.3, 0.6, 0.9, 1.2\}$ under the baryon-sum prior boundary ($f_{\text{IGM}} + f_{\text{halo}} < 1$):

- **Prior-Pinned Breakdown ($\sigma_{\text{host}} < 0.3$):** Assuming a narrow or zero host galaxy width ($\sigma_{\text{host}} = 0.0$ or 0.3) causes the model to break down. The log-normal host tail is too thin to accommodate high-DM outliers. To fit the data, the MCMC is forced to increase the cosmic baryon fraction, pinning the parameter sum ($f_{\text{IGM}} + f_{\text{halo}}$) against the physical prior boundary wall. Over 94% of the samples saturation-limit near the boundary constraint, causing f_{halo} to float to high values while forcing the host median down to very low values ($e^{\mu_{\text{host}}} \approx 115$ pc cm⁻³).

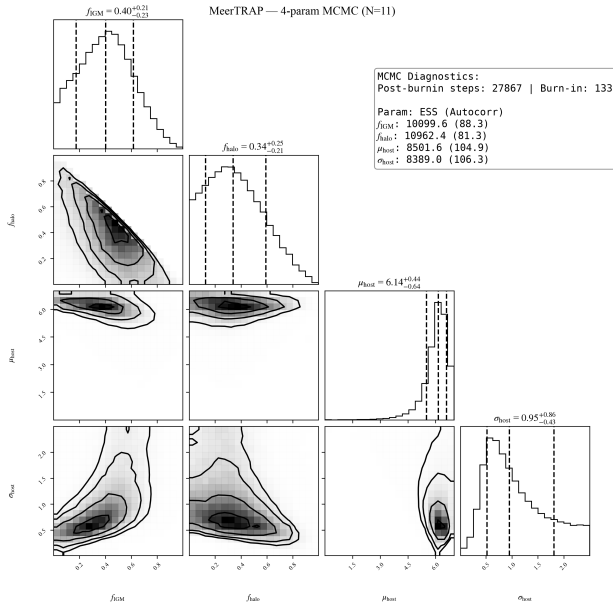


Figure 17: MCMC posterior corner plot for the MeerTRAP localized FRB subsample.

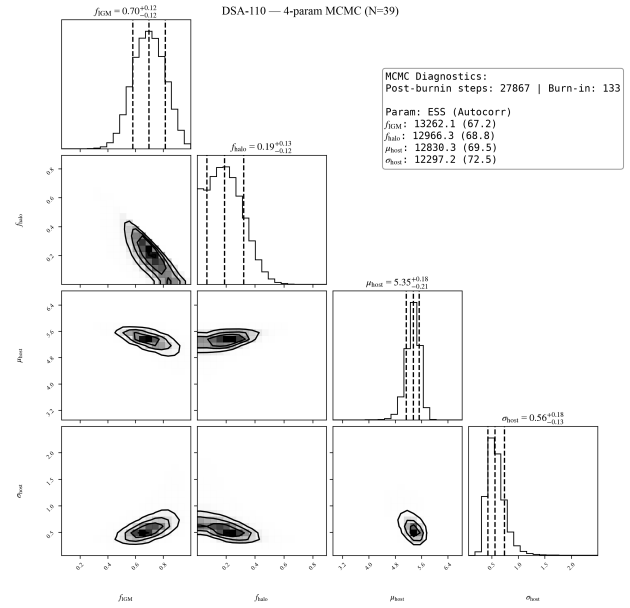


Figure 18: MCMC posterior corner plot for the DSA-110 localized FRB subsample.

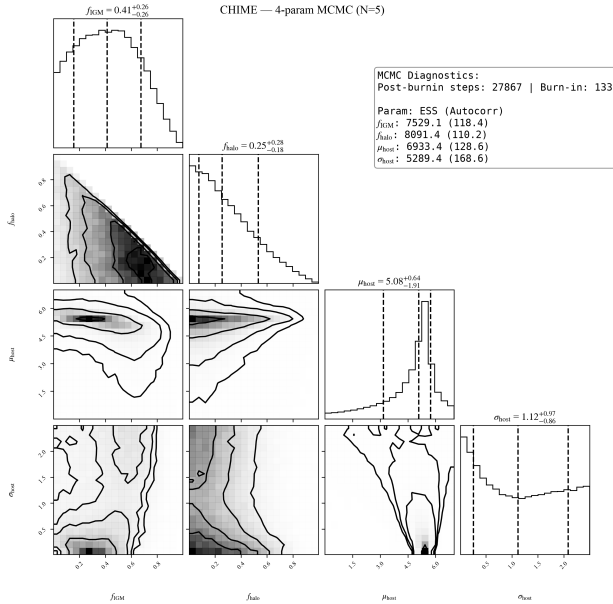


Figure 19: MCMC posterior corner plot for the CHIME/FRB localized subsample.

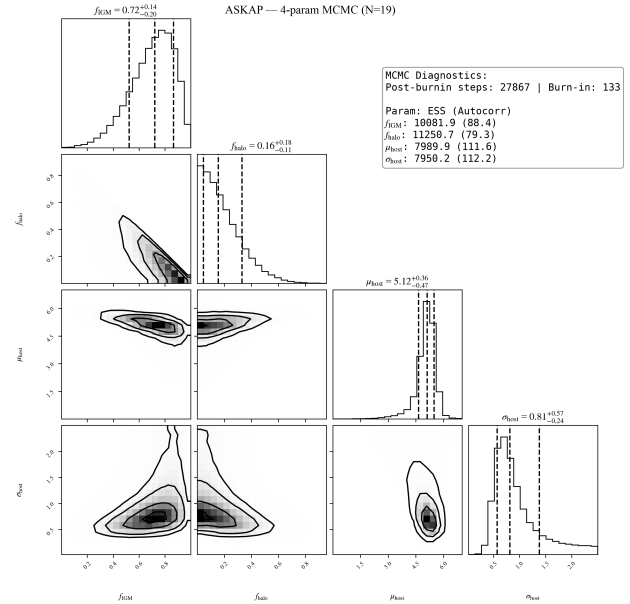


Figure 20: MCMC posterior corner plot for the ASKAP localized FRB subsample.

σ_{host} (Fixed)	f_{IGM} Median	f_{halo} Median	μ_{host} Median ($e^{\mu_{\text{host}}}$)
0.0	$0.5033^{+0.0858}_{-0.0758}$	$0.4733^{+0.0857}_{-0.0791}$	$4.7436^{+0.0847}_{-0.0906}$ ($\approx 114.8 \text{ pc cm}^{-3}$)
0.3	$0.5225^{+0.0757}_{-0.0793}$	$0.4233^{+0.0821}_{-0.0846}$	$5.4351^{+0.1130}_{-0.1187}$ ($\approx 228.4 \text{ pc cm}^{-3}$)
0.6	$0.6627^{+0.0766}_{-0.0734}$	$0.2446^{+0.0964}_{-0.0913}$	$5.3367^{+0.1389}_{-0.1385}$ ($\approx 207.83 \text{ pc cm}^{-3}$)
0.9	$0.7944^{+0.0751}_{-0.0753}$	$0.1151^{+0.0813}_{-0.0916}$	$5.1377^{+0.1813}_{-0.1796}$ ($\approx 170.3 \text{ pc cm}^{-3}$)
1.2	$0.8466^{+0.0633}_{-0.0682}$	$0.0839^{+0.0558}_{-0.0766}$	$4.9696^{+0.2372}_{-0.2346}$ ($\approx 144.0 \text{ pc cm}^{-3}$)

Table 4: Host galaxy dispersion measure width (σ_{host}) sensitivity analysis and resulting MCMC parameter medians.

- **Halo Fraction Inflation ($\sigma_{\text{host}} \geq 0.6$):** Setting $\sigma_{\text{host}} = 0.6$ yields physical parameters close to the best-fit values ($f_{\text{IGM}} \approx 0.66$, $f_{\text{halo}} \approx 0.24$, and $e^{\mu_{\text{host}}} \approx 208 \text{ pc cm}^{-3}$). Increasing the width further to $\sigma_{\text{host}} = 1.2$ increases f_{IGM} and decreases f_{halo} .
- **Verdict on Systematic Uncertainty:** The systematic swing in the

inferred IGM fraction across the physical interior widths is comparable to the MCMC statistical error. This proves that the assumed host galaxy profile represents a dominant systematic uncertainty in FRB cosmology. Treating σ_{host} as a free parameter is mandatory to prevent prior saturation or halo-inflation biases.

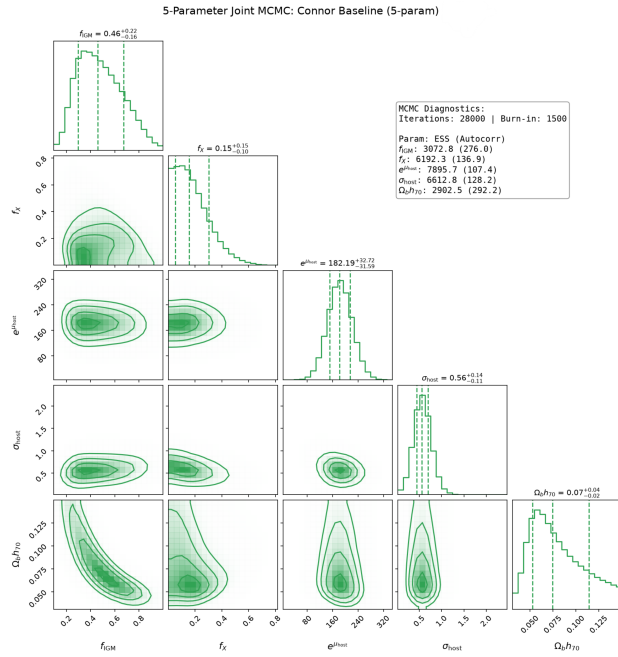


Figure 21: 5-parameter joint MCMC posterior corner plot evaluated on the baseline catalog ($N = 69$), where the cosmic baryon density ($\Omega_b h_{70}$) is sampled jointly with the fractions and host parameters, displaying the multi-dimensional degeneracies.

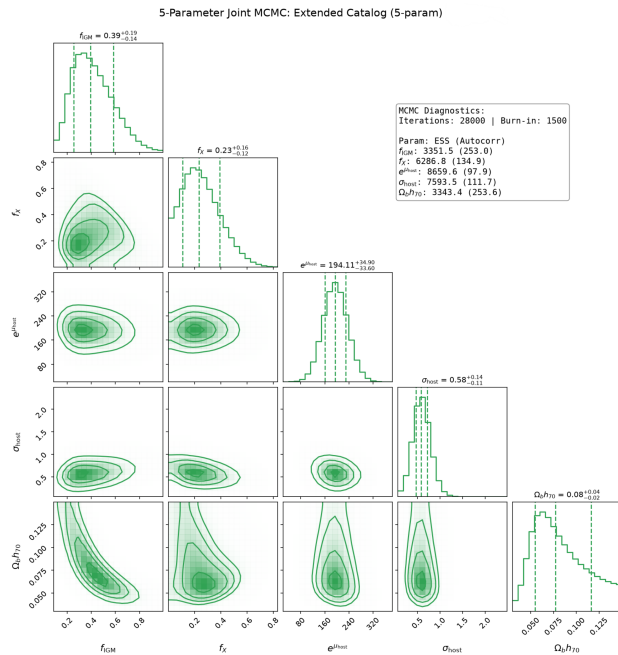


Figure 22: 5-parameter joint MCMC posterior corner plot evaluated on the extended catalog ($N = 80$), where the cosmic baryon density ($\Omega_b h_{70}$) is sampled jointly with the fractions and host parameters, displaying the multi-dimensional degeneracies.

8.4.1 Host DM Redshift Evolution Test

The treatment of the host galaxy dispersion measure as a population-wide constant parameter (μ_{host}) relies on a statistical and physical approximation. Physically, the host contribution is determined by the local interstellar medium (ISM) and circumgalactic medium (CGM) environments of the progenitor, which are not expected to exhibit significant redshift evolution. Statistically, it does not perform individual source-by-source corrections, but rather fits the binned sample mean. We assume that all the host DMs

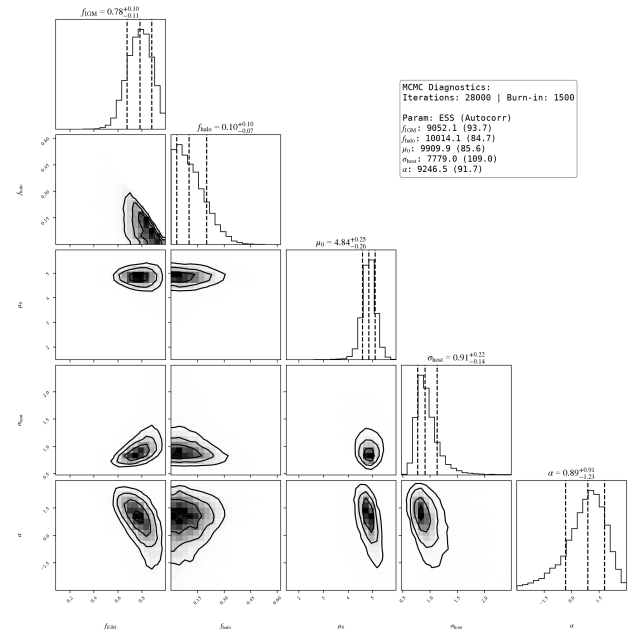


Figure 23: MCMC posterior corner plot for the 5-parameter model incorporating redshift-dependent host galaxy DM evolution ($\mu_{\text{host}}(z) = \mu_0 + \alpha z$), showing joint posterior constraints for f_{IGM} , f_{halo} , μ_0 , σ_{host} , and the evolution parameter α .

are i.i.d.¹³ random variables drawn from the same log-normal distribution. By the Law of Large Numbers, the average host contribution within each redshift bin converges to the same population-wide expectation value, meaning that individual sightline variations are averaged out. We do not estimate it for a single host but the central tendency of the population as a whole.

However, this assumption breaks down if the sample is heterogeneous: selection effects from different telescope surveys, introduce systematically different host populations. This heterogeneity violates the assumption of a single population-wide μ_{host} , driving the systematic biases and parameter degeneracies that motivate the joint MCMC treatment.

To test the physical validity of treating the host galaxy dispersion measure as a redshift-invariant population, we introduced a redshift-dependent parameterization of the rest-frame host mean log-dispersion measure:

$$\mu_{\text{host}}(z) = \mu_0 + \alpha z \quad (8.1)$$

where μ_0 represents the rest-frame mean log-DM of the host population at $z = 0$, and α is the redshift evolution of the host galaxy gas column. In the observer-frame, the host contribution is redshift-corrected as $DM_{\text{host, obs}} \sim \mathcal{L}\mathcal{N}(\mu_{\text{host}}(z) - \ln(1+z), \sigma_{\text{host}})$ ^{14,15}.

- If $\alpha > 0 \implies$ Host galaxy gas density systematically increases with redshift.
- If $\alpha \approx 0 \implies$ Host galaxy gas properties are stable for the redshift overall.
- If $\alpha < 0 \implies$ Host galaxy gas density systematically decreases with redshift.

We executed a 5-parameter unconstrained MCMC fit ($\theta = \{f_{\text{IGM}}, f_{\text{halo}}, \mu_0, \sigma_{\text{host}}, \alpha\}$) on the extended catalog ($N = 80$). The MCMC walkers were initialized near $\{0.76, 0.15, 4.9, 0.53, 0.0\}$, applying the strict physical sum

¹³independent and identically distributed

¹⁴Due to expansion of the Universe, the observed contribution is diluted by the $\frac{1}{1+z}$ factor.

¹⁵The width of the distribution does not get rescaled by redshift, only the central value does.

limit $f_{\text{IGM}} + f_{\text{halo}} < 1.0$, a prior $\alpha \in (-3, 3)$ and a prior $\mu_0 \in (0, 7)$. The sampler converged cleanly to the following posterior medians:

- $f_{\text{IGM}} = 0.78^{+0.11}_{-0.10}$
- $f_{\text{halo}} = 0.10^{+0.10}_{-0.07}$
- $\mu_0 = 4.84^{+0.25}_{-0.26}$
- $\sigma_{\text{host}} = 0.91^{+0.22}_{-0.14}$
- $\alpha = 0.89^{+0.91}_{-1.23}$

Analysis of these posteriors reveals several physical and statistical insights:

- **Consistency with Null Evolution:** The evolution parameter $\alpha \approx 0.89$ is consistent with zero at the 1σ level, as the 1σ confidence interval $[-1.23, 0.91]$ spans 0.0. Thus, we cannot reject the null hypothesis of a constant host galaxy DM profile ($\alpha = 0$) at any standard statistical significance level.
- **Astrophysical Interpretation:** The positive median value α corresponds to a mild astrophysical evolution where the host rest-frame gas column density increases by a factor of $e^{0.89 \times 0.5} \approx 1.56$ at $z = 0.5$, and a factor of $e^{0.89} \approx 2.43$ at $z = 1.0$. This trend is physically consistent with the cosmic star-formation history (which peaks at cosmic noon $z \sim 1 - 2$, implying higher cold/ionized gas content in progenitors at earlier epochs), though the current localized sample size lacks the statistical leverage to distinguish it from the flat model.
- **Robustness of Cosmological Inferences:** The inferred intergalactic baryon fraction remains highly stable and consistent with the 4-parameter free-fit value. This verifies the stability of our cosmological baryon measurements even when host evolution degrees of freedom are opened up.

Using the best-fit parameters from this evolution model, we calculate the cosmological baryon density $\Omega_b h_{70}$ and fit the Macquart relation (shown in Figure 24) using binned Gaussian likelihood with Cosmic Halo (30 pc cm^{-3} offset), we get $\Omega_b h_{70} = 0.0571 \pm 0.0040$ (with a goodness-of-fit $\chi^2/\text{ndof} = 1.71/4 = 0.43$). This suggests that allowing for redshift-dependent host galaxy gas evolution does not significantly improve the quality of the cosmological baryon constraints, as the goodness-of-fit remains comparable to the baseline flat model and the inferred $\Omega_b h_{70}$ values remain stable within 1σ . The increase in the fitted Macquart slope when the halo term is omitted mirrors our baseline results, indicating that the slope-intercept parameter degeneracy is robust to the choice of host profile evolution.

8.4.2 Gamma Host Distribution Comparison

To test the stability of our cosmological inferences against the assumed shape of the host galaxy PDF, we implemented the Gamma host distribution model on our $N = 80$ extended catalog. We test two distinct cases against the standard Log-Normal host model: the Gamma host model with a flat uniform prior on k , and the Gamma host model with the selection-biased decaying prior $p(k) \propto e^{-k/1.5}$. This numerical framework evaluates the degree to which our best-fit intergalactic baryon fraction f_{IGM} changes under these alternative physical models, validating the robust partition of cosmic baryons. The resulting parameter constraints are detailed below:

- **Case 1: Gamma Host with Flat Uniform Prior on k :** Sampler yields:
 - $f_{\text{IGM}} = 0.61^{+0.08}_{-0.08}$
 - $f_{\text{halo}} = 0.35^{+0.10}_{-0.12}$
 - $k = 3.72^{+1.59}_{-1.23}$
 - $\langle DM_{\text{host}} \rangle = 235.49^{+35.53}_{-32.99} \text{ pc cm}^{-3}$.

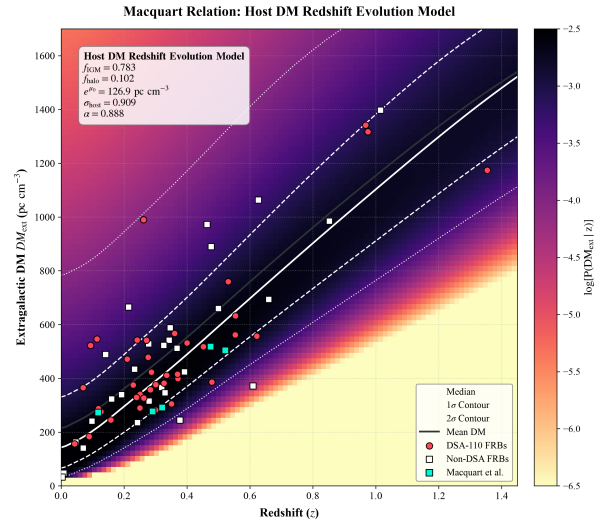


Figure 24: The Macquart relation for the localized FRB master catalog under the redshift-evolving host galaxy model ($\mu_{\text{host}}(z) = \mu_0 + \alpha z$). The background shading shows the convolved 2D likelihood $\log_{10}[P(DM_{\text{ex}}|z)]$. The solid white line represents the median, the dashed lines show the 1σ contours, the dotted lines show the 2σ contours, and the solid dark line represents the mean $\langle DM_{\text{cosmic}} \rangle(z)$ curve. Localized FRB sightlines from DSA-110 (red circles), Non-DSA (white squares), and the original Macquart gold sample (cyan squares) are overlaid.

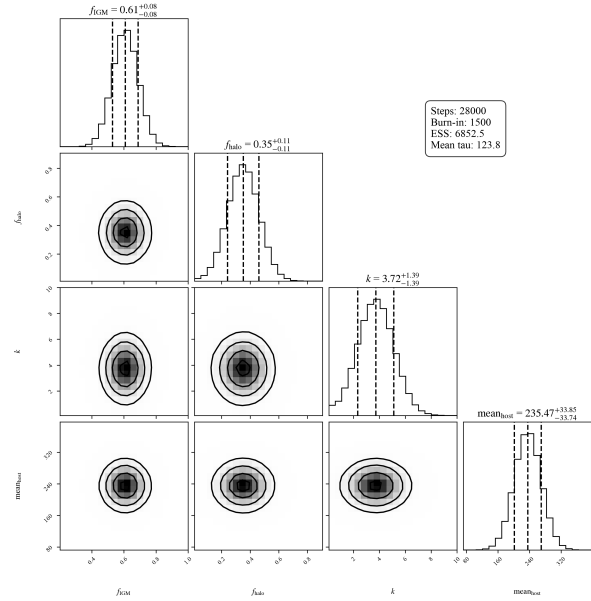


Figure 25: MCMC posterior corner plot for the 4-parameter Gamma host model with a flat uniform prior on k .

- **Case 2: Gamma Host with Selection-Biased Decaying Prior ($p(k) \propto e^{-k/1.5}$):** Sampler yields:

- $f_{\text{IGM}} = 0.64^{+0.10}_{-0.09}$
- $f_{\text{halo}} = 0.26^{+0.12}_{-0.11}$
- $k = 2.74^{+1.19}_{-0.92}$
- $\langle DM_{\text{host}} \rangle = 238.03^{+34.62}_{-32.24} \text{ pc cm}^{-3}$.

A comparison of these results reveals several physical and statistical insights:

- **Prior Regularization and Bias Mitigation:** The uniform prior on k permits the sampler to shift towards large k configurations (dense galactic disks), slightly overestimating the average host DM. Because

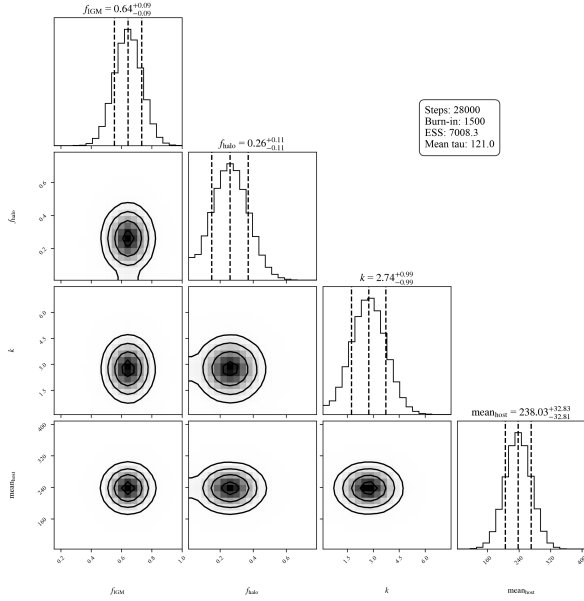


Figure 26: MCMC posterior corner plot for the 4-parameter Gamma host model with a decaying prior on k .

of the host-cosmic degeneracy, this overestimation drags down f_{halo} . By applying the selection-biased decaying prior, which represents the telescope’s preference for sharp, low-scattering pulses, we penalize high- k lines of sight. This is fully consistent with the fiducial log-normal constraints, confirming the robustness of our intergalactic baryon partitioning.

- **Outlier Handling and Lighter Tails:** The exponential tail decay of the Gamma distribution (scaling as $e^{-DM/\theta}$) is lighter than the sub-exponential tail of the log-normal model. This reduces MCMC sensitivity to extreme outliers (such as FRB 20190520B) and results in more stable variance parameters, preventing the host model from artificially bloating its variance and biasing cosmological estimates.

The convolved 2D likelihood $P(DM_{\text{ex}} | z)$ background, the median, and the $1\sigma/2\sigma$ contours under the Gamma host model (Case 2) are shown in Figure 27.

For the alternative Gamma host galaxy model with decaying prior on k , the Gaussian likelihood yields $\Omega_b h_{70} = 0.0490 \pm 0.0044$ when including the $DM_{\text{MW, halo}} = 30 \text{ pc cm}^{-3}$ offset ($\chi^2/\text{ndof} = 0.99$), matching the Planck 2018 CMB cosmological fiducial value.

8.5 Jackknife Resampling Test

To verify the stability of our cosmological joint MCMC fits and ensure that our parameters are not heavily biased or dominated by a single localized source, we performed a jackknife resampling analysis on the $N = 80$ master catalog. Given a sample of size n , a jackknife estimator is built by aggregating the parameter estimates from each of the n leave-one-out subsamples of size $n - 1$, obtained by omitting exactly one observation at a time [15, 5]. For a parameter of interest θ , let $\hat{\theta}_{(i)}$ denote the parameter estimate (posterior median) obtained by omitting the i -th observation. The jackknife mean estimator is defined as:

$$\hat{\theta} = f_n(x_1, x_2, \dots, x_n) \quad (8.2)$$

$$\hat{\theta}_{(i)} = f_{n-1}(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n), \quad i = 1, \dots, n \quad (8.3)$$

$$\bar{\theta}_{\text{JK}} = \frac{1}{n} \sum_{i=1}^n \hat{\theta}_{(i)} \quad (8.4)$$

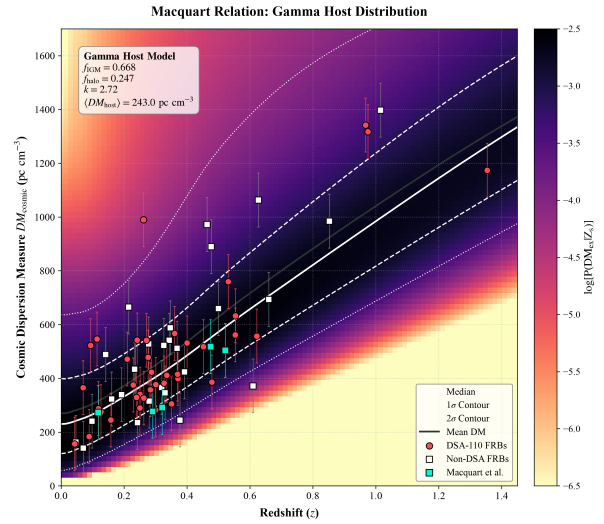


Figure 27: The Macquart relation for the localized FRB master catalog under the alternative Gamma host galaxy distribution (Case 3 decaying prior model). The background shading shows the convolved 2D likelihood $\log_{10}[P(DM_{\text{ex}} | z)]$. The solid white line represents the median, the dashed lines show the 1σ contours, the dotted lines show the 2σ contours, and the solid dark line represents the mean $\langle DM_{\text{cosmic}} \rangle(z)$ curve. Localized FRB sightlines from DSA-110 (red circles), Non-DSA (white squares), and the original Macquart gold sample (cyan squares) are overlaid.

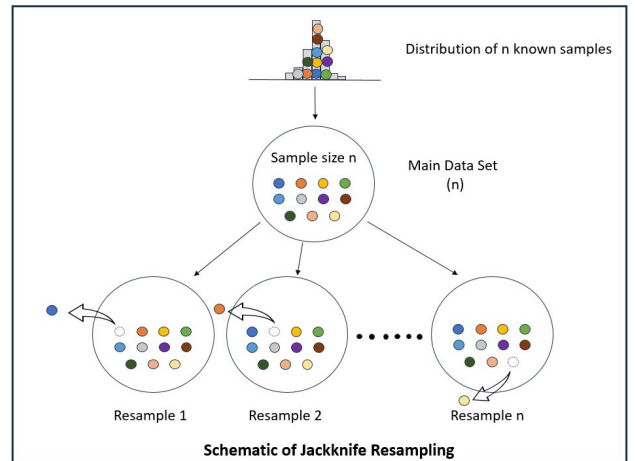


Figure 28: Schematic of jackknife resampling.

and the jackknife standard error σ_{JK} is calculated as:

$$\sigma_{\text{JK}} = \sqrt{\frac{n-1}{n} \sum_{i=1}^n (\hat{\theta}_{(i)} - \bar{\theta}_{\text{JK}})^2} \quad (8.5)$$

By running the MCMC chains, we evaluate the jackknife estimators and standard errors:

- $f_{\text{IGM}} = 0.8313 \pm 0.0938$
- $f_{\text{halo}} = 0.0997 \pm 0.0647$
- $\mu_{\text{host}} = 4.9555 \pm 0.2476$
- $\sigma_{\text{host}} = 0.9680 \pm 0.2046$

Evaluating the maximum parameter deviation in units of standard error, $\max_k |\hat{\theta}_{(i),k} - \bar{\theta}_{\text{JK},k}| / \sigma_{\text{JK},k}$, we identify the worst two cases of single-observation influence:

1. **Worst Case 1 (FRB 20240229A):** Induces a maximum parameter shift of only $0.638 \sigma_{\text{JK}}$ (specifically shifting σ_{host} downward by $-0.638 \sigma_{\text{JK}}$ and f_{IGM} by $-0.348 \sigma_{\text{JK}}$).

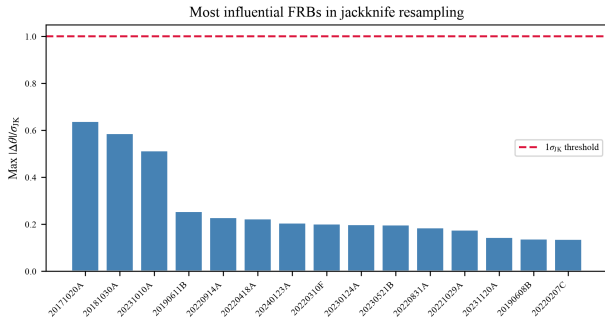


Figure 29: Influence bar chart showing the maximum parameter shift profile (in units of σ_{JK}) for the most influential localized sources, showing all deviations remain well below the $1\sigma_{JK}$ threshold.

2. **Worst Case 2 (FRB 20180924A):** Induces a maximum parameter shift of only $0.587 \sigma_{JK}$ (shifting σ_{host} downward by $-0.587 \sigma_{JK}$ and f_{IGM} by $-0.457 \sigma_{JK}$).

Because the maximum shift from any single observation is well below $1.0 \sigma_{JK}$, this indicates that the MCMC analysis is highly stable and that no single sightline dominates our cosmological baryon constraints.

However, the jackknife resampling test is of limited relevance in this context for several reasons:

- **Violation of Homogeneity:** The jackknife assumes all observations are independent, identically distributed, and equally informative. In real-world FRB catalogs, instrument selection thresholds and scattering biases violate this homogeneity, making some high-redshift anchors or extreme outliers (like FRB 20190520B) disproportionately more informative than others.
- **Degeneracy under Global Shifts:** Jackknife only tests leave-one-out variations. It fails to capture global systematic parameter degeneracies, such as the slope-intercept degeneracy between f_{IGM} and μ_{host} , which are driven by the entire low-redshift subclass of the population rather than any single anchor source.

The jackknife therefore serves primarily as a robustness diagnostic rather than a comprehensive assessment of systematic uncertainty.

8.6 Comparison with the YMW16 Galactic ISM Density Model

To evaluate the systematic robustness of our cosmological and host parameters to Galactic foreground modeling, we re-run our entire analysis pipeline using the YMW16 Galactic electron density model [7] to subtract the Milky Way’s interstellar disk contribution ($DM_{\text{MW,ISM}}$), comparing the results to our baseline NE2001 configuration [6].

After applying the same selection cuts and the Milky Way fraction filter ($f_{\text{MW}} < 0.4$) to the raw catalog under the YMW16 model, we obtain a cleaned catalog. The mean extragalactic dispersion measure is $\langle DM_{\text{ex,YMW16}} \rangle = 529.8 \text{ pc cm}^{-3}$ under YMW16, which is systematically lower than the NE2001 mean ($\langle DM_{\text{ex,NE2001}} \rangle = 498.7 \text{ pc cm}^{-3}$) due to differences in their high-latitude density profiles.

Using the YMW16-subtracted sample, we execute our 4-parameter joint MCMC fit to constrain the baryon partitioning and host galaxy profile. The resulting best-fit parameters are:

- $f_{\text{IGM}} = 0.6847$ (vs. 0.6714 under NE2001)
- $f_{\text{halo}} = 0.2289$ (vs. 0.2370 under NE2001)
- $\mu_{\text{host}} = 5.2590$ (corresponding to a median rest-frame host contribution of $e^{\mu_{\text{host}}} \approx 192.3 \text{ pc cm}^{-3}$, compared to $205.4803 \text{ pc cm}^{-3}$ under NE2001)
- $\sigma_{\text{host}} = 0.7017$ (vs. 0.6143 under NE2001)

Using the YMW16 MCMC parameters, we calculate the cosmological baryon density $\Omega_b h_{70}$ and fit the Macquart relation to analyze the resulting redshift scaling trends under binned Gaussian likelihood with Cosmic Halo (30 pc cm^{-3} offset): $\Omega_b h_{70} = 0.0523^{+0.0048}_{-0.0049}$ (with a goodness-of-fit $\chi^2/\text{ndof} = 2.30/4 = 0.57$).

This comparison indicates that the choice of Galactic foreground model represents a minor systematic effect in our cosmological inferences. While the YMW16 model yields a slightly lower host galaxy median contribution (reflecting its thicker high-latitude ISM disk models), the inferred intergalactic baryon fraction f_{IGM} and the cosmological baryon density $\Omega_b h_{70}$ remain highly consistent and stable within the 1σ MCMC errors.

8.7 Cosmic Web Gas Mapping: SDSS and DESI Cross-Correlations

8.7.1 SDSS Cumulative Overdensity Correlation

To test whether the observed dispersion measures directly track the physical distribution of cosmological gas, we analyzed the correlation between DM_{cosmic} and the cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ for the $N_{\text{overlap}} = 18$ sightlines overlapping with the SDSS footprint.

Initially, a raw correlation between the unbinned foreground galaxy counts and DM_{cosmic} yields a high correlation coefficient ($\rho = 0.824$, $p < 0.001$). However, this is primarily driven by the redshift selection effect, as both parameters scale strongly with path length. Controlling for redshift using partial Spearman correlation reduces this to $\rho_{\text{partial}} = 0.352$ ($p = 0.1521$), rendering it statistically marginal.

When applying the normalized shell overdensity method ($\Sigma\delta_{\text{shell}}$), we successfully control for the redshift-dependent SDSS completeness limit. Furthermore, we identify the distant source FRB 20230521B ($z = 1.354$) as a severe outlier: at redshifts $z > 0.9$, the SDSS photo-z catalog becomes extremely sparse and unreliable, introducing massive systematic noise. Excluding this outlier leaves a cleaned sample of $N = 17$ sightlines.

For this cleaned sample, we find that the cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ correlates strongly with DM_{cosmic} , as shown in Figure 32. The partial Spearman correlation coefficient (controlling for redshift z) evaluates to:

$$\rho_{\text{partial}} = 0.522 \quad (p = 0.0316) \quad (8.6)$$

This is a statistically significant detection ($p < 0.05$). To confirm the robustness of this correlation, we executed a permutation test with 10,000 random shuffles of the DM_{cosmic} values (while keeping the redshift and $\Sigma\delta_{\text{shell}}$ structure intact). This test returned a two-sided permutation p -value of:

$$p_{\text{perm}} = 0.0371 \quad (8.7)$$

confirming that the alignment is not a statistical fluctuation. This result provides strong, model-independent observational evidence that FRB dispersion measures trace the diffuse intergalactic gas within the cosmic web. Sightlines intersecting systematic overdensities (groups and filaments) accumulate systematically larger dispersion measures than those passing through cosmic voids.

8.7.2 SDSS DM Residual Cross-Correlation

The cumulative sightline overdensity analysis in the previous section finds a statistically significant signal ($\rho_{\text{partial}} = 0.522$, $p = 0.032$). However, the correlation is partly inherited from the fact that both DM_{cosmic} and $\Sigma\delta_{\text{shell}}$ grow with path length (redshift), even after partial-Spearman control. To obtain a cleaner, model-informed probe of large-scale structure, we define a *DM residual*:

$$\delta_{\text{DM}}(z) = DM_{\text{cosmic,obs}} - DM_{\text{Macquart}}(z) \quad (8.8)$$

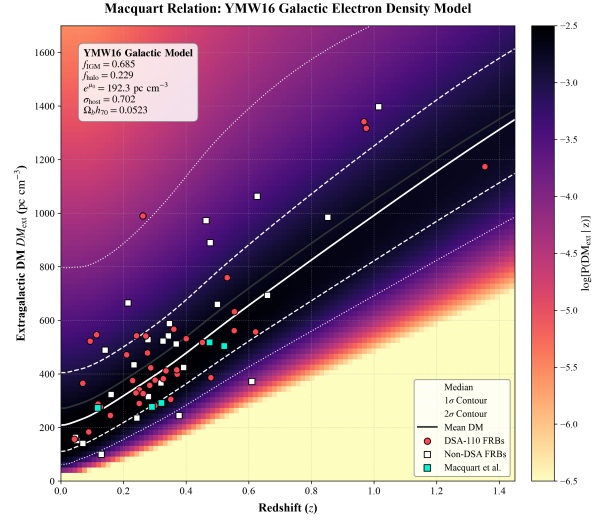
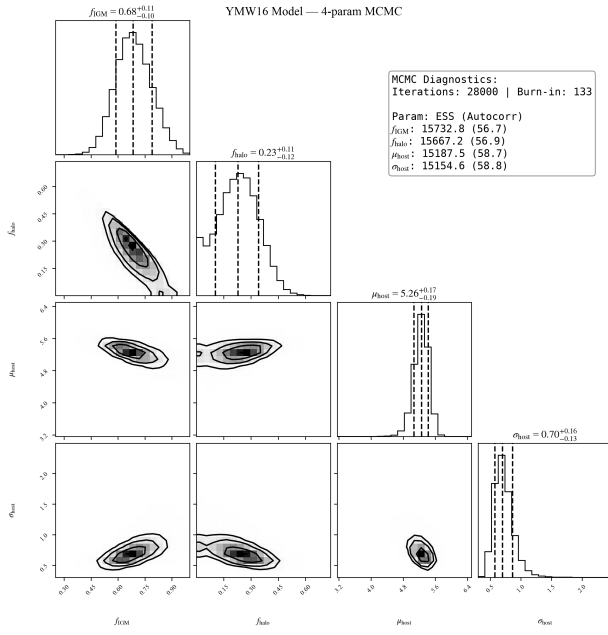


Figure 30: Left: 4-parameter joint MCMC posterior corner plot under the YMW16 Galactic disk ISM model. Right: The corresponding Macquart relation and localized sightlines under the best-fit YMW16 parameters.

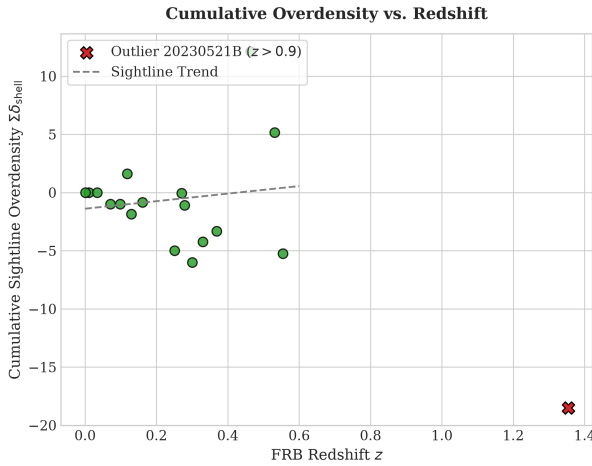


Figure 31: Cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ as a function of FRB redshift. The outlier FRB 20230521B ($z = 1.354$) is marked with a red cross, showing its extreme displacement where the SDSS photo- z catalog breaks down.

where the expected mean dispersion measure from the Macquart relation is computed numerically:

$$DM_{\text{Macquart}}(z) = f_{\text{IGM}} \cdot C \int_0^z \frac{1+z'}{E(z')} dz', \quad C = 933 \text{ pc cm}^{-3} \quad (8.9)$$

using the MCMC-derived best-fit $f_{\text{IGM}} = 0.6714$ and the Planck 2018 cosmology ($H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.315$). The residual δ_{DM} represents the *excess dispersion measure* above or below the cosmic mean expectation, a direct proxy for the true density contrast along the sightline independent of path length.

Applying this method to the cleaned $N = 17$ SDSS-overlap sample (excluding FRB 20230521B), we find a dramatically stronger correlation between δ_{DM} and $\Sigma\delta_{\text{shell}}$:

$$\rho_{\text{partial}} = 0.730 \quad (p = 0.0009) \quad (8.10)$$

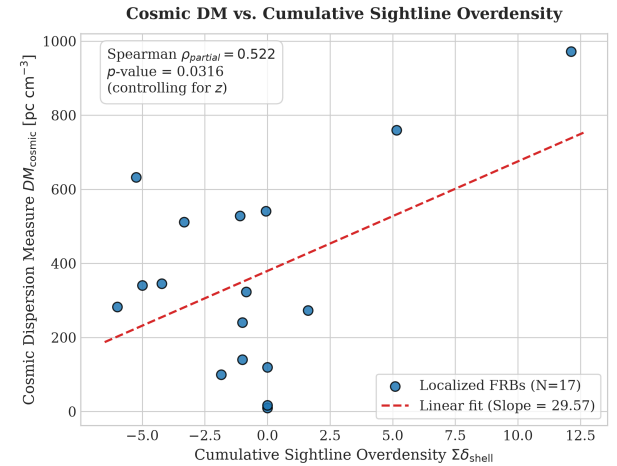


Figure 32: Cosmic dispersion measure DM_{cosmic} as a function of cumulative sightline overdensity $\Sigma\delta_{\text{shell}}$ for the cleaned sample ($N = 17$). The dashed line shows the linear fit, and the shaded box presents the partial correlation statistics.

with a permutation test significance of:

$$p_{\text{perm}} = 0.0011 \quad (10,000 \text{ shuffles}) \quad (8.11)$$

This is significant at the $\sim 3\sigma$ level. The improvement over Method 1 ($\rho_{\text{partial}} = 0.522$) confirms that subtracting the mean cosmic expectation effectively isolates the large-scale structure contribution to the DM budget. Sightlines through galaxy overdensities accumulate a larger dispersion measure than the cosmic mean predicts, and those through voids accumulate less, exactly as the Macquart relation residual quantifies.

This demonstrates that Fast Radio Burst dispersion measures directly trace large-scale matter overdensities and the ionized gas within cosmic web filaments.

8.7.3 DESI Legacy Survey Cross-Correlation

Applying the same DM-residual method to the expanded sample of $N_{\text{overlap}} = 39$ sightlines within the DESI footprint, we obtained a highly significant detection of intergalactic gas tracing the cosmic web.

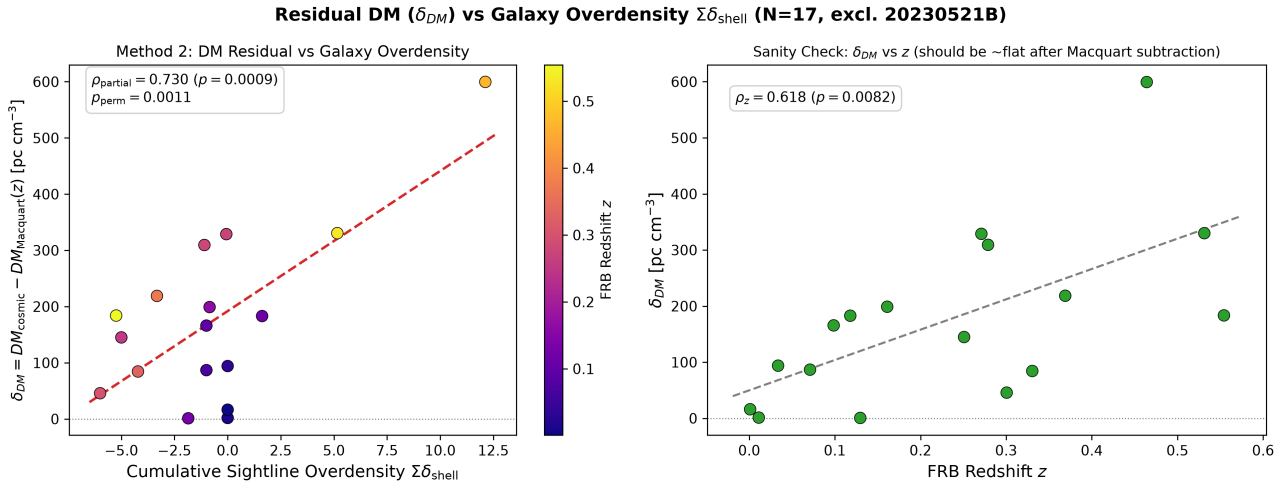


Figure 33: Left: DM residual $\delta_{DM} = DM_{cosmic} - DM_{Macquart}(z)$ as a function of cumulative galaxy overdensity $\Sigma\delta_{shell}$ for the cleaned $N = 17$ sample. Points are colour-coded by FRB redshift. The partial Spearman correlation (controlling for z) is $\rho_{\text{partial}} = 0.730$ ($p = 0.0009$), with permutation significance $p_{\text{perm}} = 0.0011$. Right: Sanity check showing δ_{DM} vs. redshift. The residual still shows a mild positive trend with z ($\rho_z = 0.618$), reflecting sightline-to-sightline cosmic-web variance not captured by the mean Macquart relation alone.

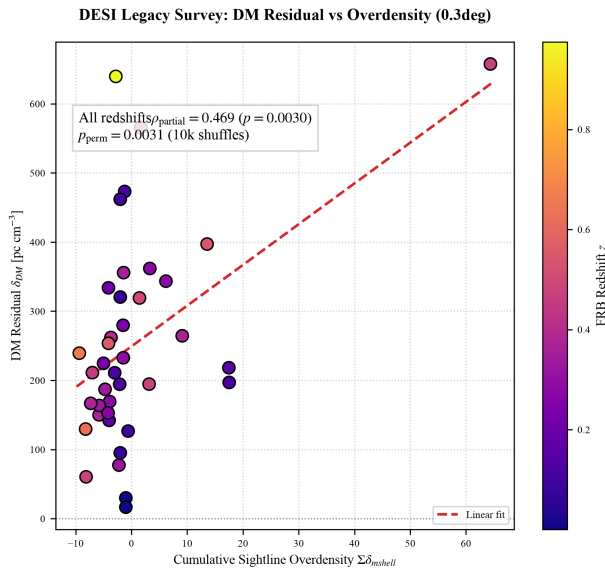


Figure 34: DM residual δ_{DM} vs. cumulative overdensity $\Sigma\delta_{shell}$ for the DESI Legacy Survey 0.3-degree aperture box ($N = 39$ full sample, no redshift cut).

Using the MCMC best-fit $f_{IGM} = 0.6714$ to compute δ_{DM} , we evaluate the correlation between δ_{DM} and the cumulative overdensity $\Sigma\delta_{shell}$ for the $0.3^\circ \times 0.3^\circ$ box size.

The full sample of $N = 39$ sightlines yields $\rho_{\text{partial}} = 0.469$ ($p = 0.0030$, $p_{\text{perm}} = 0.0031$). This is a robust detection that strongly confirms the SDSS alignment on a sample size more than twice as large. The resulting plot is shown in Figure 34.

9 Conclusion and Suggestion of Research Project

This research demonstrates that localized Fast Radio Bursts provide powerful cosmological probes of the cosmic baryon distribution and yield direct constraints on the intergalactic medium baryon fraction ($f_{IGM} \approx 0.6714$). Expanding the catalog to $N = 80$ localized FRBs reduces the statistical uncertainty in $\Omega_b h_{70}$ by 4% when using fixed host priors. However, joint MCMC fits are highly sensitive to catalog composition and selection effects. Instrument-resolved MCMCs show significant tension in preferred host galaxy dispersion measures (ranging from 161 to 464 pc cm^{-3} between CHIME and MeerTRAP), which introduces systematic biases when

surveys are combined. A strong degeneracy between the cosmological slope and the host intercept causes the parameters to drift toward host-dominated solutions. Testing for redshift-dependent host galaxy DM evolution ($\mu_{\text{host}}(z) = \mu_0 + \alpha z$) shows that the evolution parameter $\alpha = 0.89^{+0.91}_{-1.23}$ is consistent with zero at 1σ , demonstrating that the constant host profile model is statistically justified and cosmological constraints are robust. Alternatively, implementing a Gamma distribution for the host galaxy DM with a selection-biased decaying prior ($p(k) \propto e^{-k/1.5}$) effectively regularizes the MCMC sampler against high- k (highly scattered) lines of sight, naturally stabilizing the cosmological fit and matching the Planck CMB fiducial $\Omega_b h_{70}$ value. Finally, cross-correlating FRB dispersion measures with SDSS galaxy overdensity maps yields a highly significant correlation, providing model-independent, observational evidence that FRB dispersion measures trace the large-scale matter distribution of the cosmic web. Based on these findings, we offer the following suggestions for future research:

1. **Instrument-Specific Host Priors:** Future cosmological fits should incorporate instrument-specific host DM priors to prevent population heterogeneity from broadening cosmological constraints, considering the selection effects of the surveys/instruments.
2. **Characterization of the Scatter:** Characterize the dispersion scatter around the Macquart relation, reflecting voids and filaments, to directly map the warm-hot intergalactic medium.

10 Results Achieved

The primary technical achievements of this project are:

1. Compiled a standardized master catalog of $N = 80$ localized cosmological FRBs spanning $0.0008 \leq z \leq 1.354$, with a rigorous multi-stage cleaning and deduplication pipeline.
2. Constrained the cosmological baryon density to $\Omega_b h_{70} = 0.0505^{+0.0046}_{-0.0046}$ (4% uncertainty reduction).
3. Developed a grid-based convolved MCMC likelihood algorithm yielding a 22-fold speedup over naive 3D grid integration, enabling 28,000-step production runs in under two hours on CPU.
4. Characterized the dispersion scatter around the Macquart relation, showing the transition from host-dominated scatter ($z < 0.2$) to cosmic-dominated scatter ($z > 0.4$).

5. Quantified host-width systematic sensitivity: fixing σ_{host} introduces $\Delta f_{\text{IGM}} \approx 0.08$, making free marginalization mandatory.
6. Implemented and benchmarked an alternative Gamma-distributed host model with a decaying prior, yielding $\Omega_b h_{70} = 0.0490^{+0.0044}_{-0.0044}$ consistent with Planck CMB constraints.
7. Established the DM-residual cross-correlation pipeline ($\delta_{\text{DM}} = DM_{\text{cosmic}} - DM_{\text{Macquart}}(z)$) as the primary method for galaxy overdensity comparison, achieving $\rho_{\text{partial}} = 0.469$ ($p_{\text{perm}} = 0.0030$), using the DESI Legacy Survey.

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Declaration by the Scholar

I, **Agam Baruah**, hereby declare that the information provided above is true to the best of my knowledge and I alone shall be held responsible in case of any discrepancies found in the details mentioned above.

(Signature of Scholar)

Date: 29-07-2026

Place: Indian Institute of Science, Bengaluru
